Instructor Guide to ‘Learn Around the Board’

By Hans Ghezzi · *Living edition, last updated October 2026*

This guide lives in the project repository and is rebuilt every time the game changes, so it always matches the version at https://hghezzi.github.io/Learn-Around-the-Board/. Read it on the web or as a PDF. For your students, there is a short Student Guide (web · printable PDF).

New here? Start with the Quick start. **Used it before?** See What’s new.

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Quick start

You can try the game in five minutes and run your first class with your own questions the same week. It is free. You need a browser and, for your own questions, a spreadsheet; you don't need to install anything, create an account, use GitHub or set up a server.

1. **Play the demo (5 minutes).** Open https://hghezzi.github.io/Learn-Around-the-Board/ and click **Play the demo** (16S rRNA sequencing) or **Try a different subject** (Intro Statistics). Continue to the setup, choose **Solo**, a topic and **Start game →**, then answer the survey and roll a few times. Click **End game** to see the end of a session.
2. **Make your question file.** Pick one of two routes:

    - **Do it yourself:** save the Intro Statistics example to your computer (right-click the link, then **Save link as…**) and open it in Google Sheets (**File → Import → Upload**). Keep the header row and replace the rows with your own questions. Designing the input file explains every column.
    - **Let Claude draft it:** the question-writer skill interviews you about your course and writes a complete, checked file for you to review.
3. **Check it.** Download it as **Tab-separated values (.tsv)** (in Google Sheets: **File → Download**), load it in the game with **Upload** and read the file check. Fix every red item.
4. **Decide how students get the questions:** a game link to a published Google Sheet, or the file itself, optionally password-protected. See Sharing the game with students.
5. **Decide how results reach you.** By default, students download a results file and submit it on your course page. With one or two extra rows in the file, they can email it to you or send it straight to your own Google Sheet instead. See Collecting results.
6. **In class:** seat groups of 2–4 students; each group plays as one player, and up to four players can share a computer. Give students the link (or file) and the Student Guide, and tell them which session length and module to pick. A suggested 80-minute plan is in Structuring your time.

Before class: play your own file once, with a 2-player game, from start to results screen, on the kind of computer students will use. It takes about 10 minutes and catches almost every problem.

Introduction

Educational practices are constantly evolving, moving away from passive absorption towards active engagement. Growing up in a memorization-heavy system, I experienced firsthand the limitations of rote learning and the powerful impact that active, student-centered strategies can have on both motivation and long-term retention. As instructors, our primary goal is to support such deeper learning, but we should not view this as a strictly rigid or purely serious process.

This is where the potential of playful pedagogy becomes transformative. By integrating game mechanics such as competition, resource management, and immediate feedback into the curriculum, we can create a dynamic environment where students feel safe to fail and eager to try again. The game I developed, Learn Around the Board (LAB; called Science Around the Board until October 2026), leverages these principles to turn reviewing

into an engaging learning cycle. This manual will guide you through that transition, showing you how to align the dynamic energy of a board game with the rigorous learning objectives of your course.

Definitions

What is playful pedagogy?

Playful pedagogy is an instructional approach that integrates the structures and spirit of play (curiosity, experimentation and social interaction) into the learning process. In this model, teachers become facilitators rather than remaining the center of the classroom. This shift establishes a "safe space" where students can take risks, make mistakes, and learn iteratively without the immediate anxiety of grade penalties.

What is Learn Around the Board?

Learn Around the Board is a customizable, web-based, open-source educational platform developed to gamify the learning of complex topics.

- **Setup:** instructors provide a question file in tab-separated values (.tsv) format, plus optional images. One to four players share a computer and load the file, or simply open a game link the instructor shares.
- **Pre-game survey:** before starting, each player completes a short pre-assessment.
- **Moving around the board:** players take turns rolling a pair of dice and moving around a square board. Each side of the board is dedicated to one theme.
    - **Unowned tile:** the player answers one of your questions for the option to buy it.
    - **Rival's tile:** the player pays rent, and a correct answer halves it.
    - **Corner milestones:** the player must pass a short exam to capture the corner, which earns a Chaos Token. Chaos Tokens can be spent to try to steal a rival's property.
    - **Full colour group:** a player who owns every tile of a colour can add stars to charge much higher rent.
- **The end:** the game ends when only one player is left standing, or when time runs out, in which case the player with the highest net worth wins.

- **Post-game survey:** afterwards, players answer the same survey again, so instructors can track learning.

The game leverages key pedagogical frameworks, such as the BOPPPS model of lesson design, along with active learning techniques and frequent low-stakes assessments to incentivize and scaffold learning. Instructors should include explanations with each question to provide frequent learning opportunities.

While originally designed for a 4th-year undergraduate research course in Microbiology, the modular nature of the game engine works seamlessly with absolutely any topic, as the content is determined by the input file. Its versatility is limitless: I have used the exact same engine to run advanced genomics workshops and to host a RuPaul's Drag Race trivia night with my wife! The built-in *Intro Statistics* example shows the same engine with a completely different subject.

Why use 'Learn Around the Board'?

Successfully leveraging play in the classroom requires aligning activities with the learning objectives. Misalignment, requiring effort allocation towards tasks that do not move students forward in the learning process, can be counterproductive by decreasing motivation and attention. It is critical to be intentional, ensuring games align with Learning Outcomes (LOs), and that the fun does not distract from the learning. A useful strategy when brainstorming playful frameworks is to begin thinking about which verb to address in Bloom's Taxonomy.

In my experience as an instructor, TA, and facilitator I recognised that students often struggle to conceptualise complex class material, frequently performing tasks without understanding the theory behind it. Students frequently "do" tasks without "knowing", thus the playful activity I had envisioned had to address the Bloom's verb "understand", which involves gaining more knowledge through review and practice. Learn Around the Board was built to bridge this gap: it enables students to step back from the rote execution of tasks and confront the conceptual framework that supports them. LAB is suited to test the ability of students to recall and recontextualise class material in novel applications.

Learn Around the Board is best suited as a replacement for review sessions, with successful execution and student feedback for both individual or group use, and synchronously or asynchronously. Instructors can design questions spanning any range of topics and use matching pre- and post-assessments to track students' learning throughout

the game. Additionally, Learn Around the Board incorporates numerous opportunities for iterative assessment followed by scaffolded explanations, enabling learning opportunities at every stage while removing the fear of failure.

The “data sandwich”: how a session measures learning

Every session wraps the game in measurement, so you learn something about your class each time:

1. **Pre-game survey (baseline).** Each player rates its confidence on your learning objectives (0–10 sliders) and answers 10 knowledge questions drawn at random from your `survey` rows.
2. **Gameplay (the intervention).** Dozens of low-stakes questions, each followed straight away by your explanation. Getting a question wrong costs only in-game money, so mistakes are cheap and memorable.
3. **Post-game survey (growth).** The same sliders and the same 10 questions again. The end screen shows each player’s score before and after.
4. **Results file (the record).** Every answer, transaction and survey response, ready for a spreadsheet. See What the results contain.

Comparing confidence with survey scores also shows calibration: a player who rates their confidence 9/10 but scores 4/10 has a misconception worth addressing, and one that does the reverse may need encouragement.

Which mechanics exercise which skills

Game mechanic	Bloom’s level	Why
Property questions and rent defense	Remember, understand	Quick retrieval of facts and concepts, many times per session
Milestone exams	Apply, analyze	Six harder questions in a row on one theme, with little room for error
Image questions	Apply	Interpreting a plot, a gel or a code snippet rather than recalling a fact
Upgrades	Analyze (the strategy)	Deciding whether to invest or keep cash means weighing risk
Chaos challenges	Evaluate (metacognition)	A player must judge whether it knows a rival’s topic well enough to bet on it

Important links

What	Link
Game	https://hghezzi.github.io/Learn-Around-the-Board/
This guide (web / PDF)	Web version · PDF
Student Guide (web / PDF)	Web version · PDF
Example files to start from	Intro Statistics (.tsv) · 16S demo (.tsv)
Complete file format reference	format.md on GitHub
Encryption tool	https://hghezzi.github.io/Learn-Around-the-Board/encryptor.html
Question-writer skill for Claude	Download (.zip)
Results collector for Google Sheets	Apps Script code (see Collecting results)
Privacy notice	https://hghezzi.github.io/Learn-Around-the-Board/privacy.html
Source code, questions and bug reports	GitHub repository · Issues

How to use ‘Learn Around the Board’

Choosing a topic

This is probably the most straightforward portion of the guide. While Learn Around the Board was designed for a microbiology course, instructors can use the game engine for absolutely any topic. Pick a subject that is suitable for your course and that aligns with your specific learning outcomes.

Designing the input file

Learn Around the Board requires students to load a file with all questions (required) and any images that the questions reference (optional). The input file must be in Tab-Separated Values (TSV) format, which I recommend creating with Google Sheets, then exporting as `.tsv` (**File → Download → Tab-separated values**). In Excel, use **File → Save As → Text (Tab delimited) (.txt)**; the game accepts `.txt` files too, but check that accented letters and symbols such as β survived. The quickest start is to import the Intro Statistics example into a Sheet and replace its rows. Column names are case-sensitive:

Column	Required?	Description	Example
`id`	Yes	Unique ID for the row	`bio_01`
`question`	Yes	The text prompt (for `mishap` rows, the wildcard text)	What is the start codon?
`option1` – `option4`	For choice questions	Answer choices A–D	AUG
`correctIndex`	For choice questions	Position of the correct option (1–4), or a list for select-all	`1`, or `1,3`
`explanation`	Yes	Feedback shown after answering	AUG codes for methionine.
`type`	Yes	How the row is used (see below)	`property`
`bigTopic`	Recommended	Overarching topic, shown in the setup menu (blank = every topic)	Molecular biology
`module`	Recommended	Sub-topic within `bigTopic` (blank = every module)	Week 3
`theme`	For board questions	One of 4 themes per module (a side of the board)	Genetics
`subtheme`	For property questions	One of 2 property groups per theme	Translation
`imageFile`	No	Image filename (must match the uploaded file) or a web link	`diagram_a.png`
`format`	No	Answer format; blank means multiple choice	`multi`, `numeric`, `order`, `text`
`answer`	For numeric/short answer	The number, or accepted answers separated by `|`	`1500`, `beta|beta diversity`
`tolerance`	No	Allowed error for numeric answers	`0.5` or `5%`

While the table above provides a quick reference, understanding how these columns interact is the key to mastering Learn Around the Board. The game engine reads your TSV

file and dynamically constructs the board, the menus and the pre- and post-game surveys based entirely on the text you provide. For example files, see the 16S demo file, which includes questions with images, or the Intro Statistics example, which uses every question format in a non-biology subject. The complete format reference covers every rule in detail.

Board hierarchy and organization

These columns act as the blueprint for your game. They tell the engine where to place questions and how to organize the main menu.

- **bigTopic and module:** these define where your game lives in the setup menu. For example, if your bigTopic is "Microbiology" and your module is "16S Sequencing", students will select those options to start.
  - *Pro-tip:* you can use comma-separated lists here. If a question applies to several modules (e.g. "16S Sequencing, Whole Genome Sequencing"), the engine pulls it into both games automatically. A blank cell means "use in every game".
- **theme (the 4 sides):** the engine uses the first four unique themes it finds (in `property` and `milestone` rows) to build the four sides of the board. Think of a theme as a major chapter of your module. Put your rows in theme order, because the order sets the board layout.
- **subtheme (the property groups):** within each side, the engine creates two property groups of three tiles each, based on the first two subthemes of that theme. These are the specific topics students will buy and defend (e.g. "DNA Extraction" and "Library Prep"). Tiles in the first group cost $100 and tiles in the second cost $160.

Each side of the 36-tile board therefore holds a corner milestone, three tiles of its first subtheme, a core tile, three tiles of its second subtheme and a Wildcard tile. START is one of the corners.

Game logic and rules

- **type:** the most important control column. It dictates when and how a row is used in the game:
  - **property:** standard questions used when a player tries to buy or defend a regular tile.
  - **milestone:** harder, comprehensive questions used for the corner exams. Students must answer 5 out of 6 correctly to capture a milestone, so provide at least 6, and ideally 8–10, per theme.

- **core:** questions for the four core tiles (the first `core` row's subtheme names them, e.g. "UNIX" or "Statistics"), shared across the board.
- **mishap:** the wildcards, random events drawn when a player lands on a Wildcard tile. Write the event text in the `question` column, including the amount, e.g. *"Someone left the freezer open! (-$100)"* or *"Scholarship awarded! (+$150)"*. The game charges or pays exactly that amount (without an amount: +$50 if the text contains a +, otherwise −$100). Put a fun fact or practical lesson in the `explanation` column. Theme them to your subject: a history course might use *"Archive flooded! (-$100)"*. Without `mishap` rows, the game uses general built-in cards with study tips.
- **survey / confidence:** these rows bypass the board and form the pre-game and post-game assessments. Ten `survey` questions are drawn at random for each player, and the same ones are asked again after the game. `confidence` rows are statements (e.g. *"I can explain how DADA2 denoises reads"*) shown as 0–10 sliders, the same for every player.
- **config:** optional settings, not questions (see Config rows below).

How many questions? The game only refuses a file with fewer than 4 themes, or a theme with no property or no milestone questions. Everything else is a recommendation: smaller pools load and play, they just repeat sooner.

Per game	45 min	60 min	90 min
`property`, per subtheme	6	8	10
`milestone`, per theme	8	8	10
`core`	6	8	10
wildcards (`mishap`)	6	8	8

Whatever the length, write 15 `survey` questions for 1–2 players and 20 for 3–4 (each player draws 10), and 4–5 `confidence` statements, one skill each.

Question formats

Multiple choice is the default, but the `format` column unlocks other question types. Every format is simply marked right or wrong, so the game rules are the same for all of them.

Format	What to fill in	What students see	Marked correct when…
(blank) multiple choice	`option1` – `option4`, `correctIndex`	Four answer buttons	the right option is clicked
True/false	`option1` = True, `option2` = False, `correctIndex`	Two buttons	as above
`multi` (select all)	options, `correctIndex` such as `1,3`	Checkboxes and Submit	the selection matches exactly
`numeric`	`answer`, optional `tolerance`	A number box	the answer is within tolerance
`order`	options written **in the correct order**	A shuffled list with ↑/↓ arrows	the order matches exactly
`text` (short answer)	`answer` with alternatives separated by `|`	A text box	it matches an accepted answer (case, accents and punctuation are ignored, and one small typo is forgiven in answers of 8+ characters, but never in the first letter, a number or a Roman numeral)

Tips: the game already shows "Select all that apply." under multi-select questions, so vary how many options are correct (1 to 4) rather than hinting at it. State units and rounding in numeric prompts; a unit typed after the number is accepted ("12 kg" counts as 12). Keep short answers to one to three words, list every reasonable synonym, and use multiple choice when one letter turns a long answer into a different term (absorption and adsorption): the typo rule would accept it. Numbers and Roman numerals never get the typo allowance, so "Type II" is never accepted for "Type I".

Config rows (optional settings)

Rows with `type` = `config` are not questions: they set up how results reach you. Put the setting name in `id`, its value in `question`, and leave the other columns blank.

`id`	Value (in the `question` column)	Effect
`results_url`	the Web app URL of your results collector (`https://script.google.com/macros/s/…/exec`)	adds **Send results to instructor** to the end screen (see Collecting results). Any other address is ignored
`instructor_email`	your email address	adds **Email results to instructor**
`course`	e.g. BIOL 301 – Week 5	labels the results in your Sheet and in the email
`ask_names`	`yes` or `no`	whether names or student IDs must be typed before **Send**, **Email** or **Download** (the boxes are always shown); the default is yes when results are sent or emailed

Config rows apply to the whole file, whatever topic and module students pick. The file check flags links and addresses that won't work.

The assessment content

This is the actual material your students will interact with during play.

- **id:** a unique identifier for your own tracking (e.g. `WGS_001`). It appears in the exported data, which makes it much easier to match results to specific learning objectives.
- **question:** keep this clear and concise. If the prompt is too long, students may experience reading fatigue during fast-paced competitive play.
- **option1–option4:** the answer choices. The engine shuffles their order every time the question appears, so you don’t need to randomize them yourself. Avoid "all of the above" and options that refer to positions ("both A and C"), because positions change; use the `multi` format instead.
- **correctIndex:** the position (1, 2, 3 or 4) of the correct answer as written in your file.
- **explanation:** this is your pedagogical safety net. It appears immediately after a student answers, whether they were right or wrong, together with the correct answer. Use it to clarify misconceptions, give a fun fact, or reinforce the learning objective.

- **imageFile:** if a question needs a visual aid (a graph, gel image or code snippet), put the exact filename here (e.g. `gel_lane_1.png`).

Spreadsheet pitfalls

- **Cells that start with `=`, `+` or `-`** (for example a command-line flag such as `--p-sampling-depth`) are turned into formulas by Google Sheets and Excel, and export as `#NAME?`. Type an apostrophe first (`'--p-sampling-depth`), or wrap the text in backticks, then check the exported file.
- **No line breaks or tabs inside a cell.** The file is split on tabs and line breaks, so either one breaks the row.
- **Keep `type` values lowercase** (`survey`, not `Survey`): survey and confidence rows are matched exactly.
- Lines that start with `#` are ignored, so you can leave notes to yourself in the file.

Images (optional)

Images are always optional. For each question with an `imageFile`, the game looks for:

1. a file the students uploaded with the same name;
2. a full web link (`https://…`) written in the column;
3. a file in the image folder of a shared game link (see Sharing the game with students);
4. an image hosted with the game (only the demo's own figures).

If none is found, the question is shown without the image, and the start page lists the images it couldn't find. Filenames are case-sensitive (`graph.PNG` is not `graph.png`). Students upload all images at once with the **Optional: upload images** button after loading the question file.

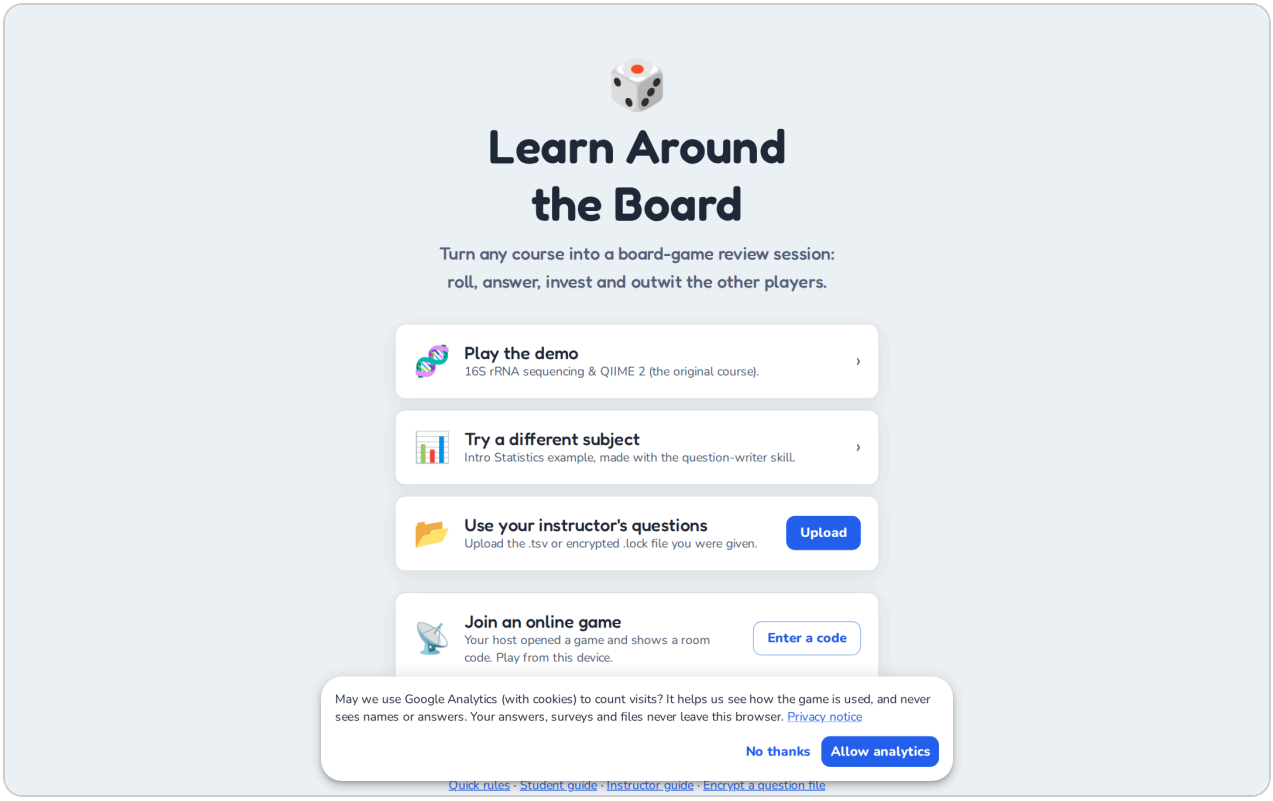
Checking your file

When a file is loaded, the game checks it and lists any problems before the session starts:

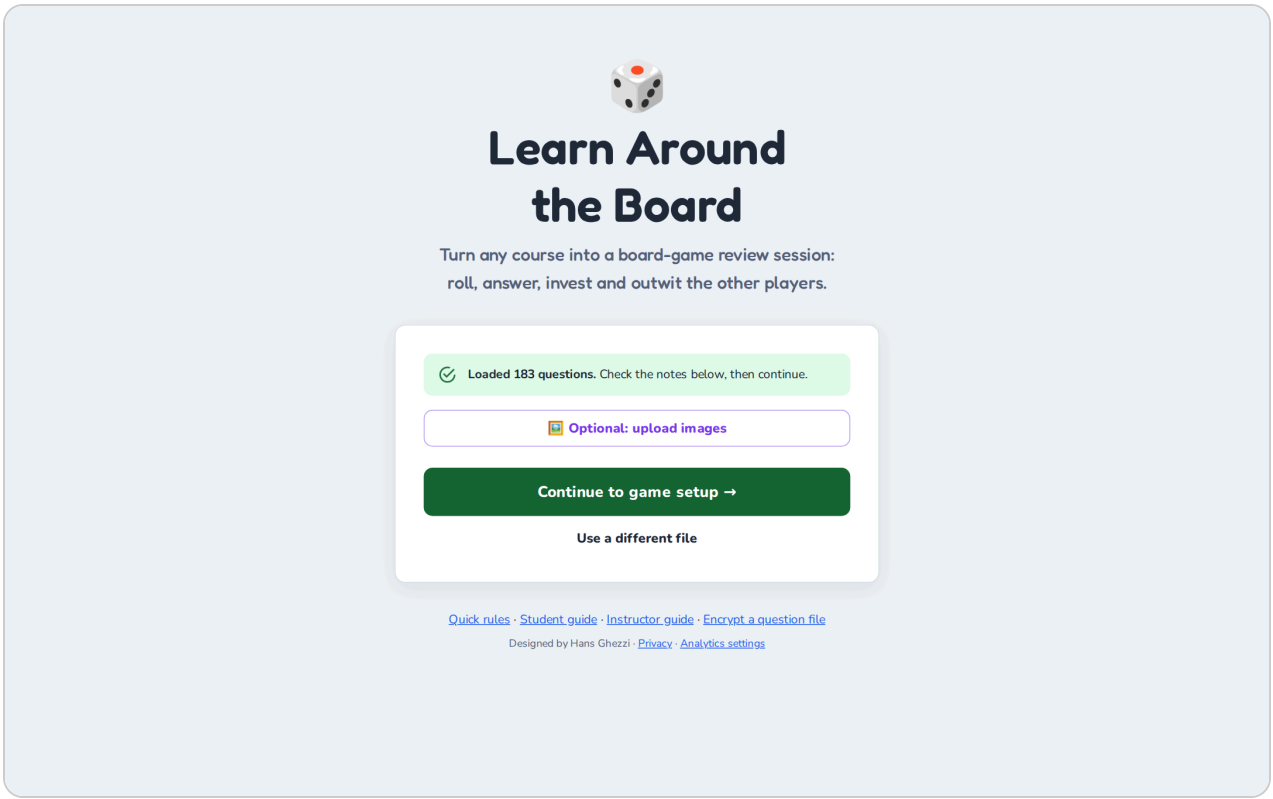
- **Red items** will break the game or make a question impossible to answer correctly. Examples: fewer than 4 themes, a `correctIndex` that doesn’t point to an option, or a theme with no milestone questions.
- **Yellow notes** (click to expand) are suggestions. Examples: too few survey questions, wildcards without an amount, or ignored rows.

- **Answer cues** are yellow notes about question quality. Test-wise students pick the longest or most detailed option, avoid options with "always" or "never", and pick the option that repeats the question's words. The checker measures how well each of these blind strategies would score on your file. For four options, chance is 25%; if "pick the longest option" would score, say, 60%, students can win without knowing the content. Fix it by making the wrong options as long and as detailed as the right one (real misconceptions, not padding), or by moving the extra detail from the correct option into the explanation.
- **Spreadsheet errors** such as `#NAME?` or `#ERROR!` are red items. Excel and Google Sheets turn text starting with `-`, `+`, `=` or `@` (for example `--p-sampling-depth`) into a formula, so the checker also warns about such cells before a spreadsheet damages them. Wrap commands in backticks. A cell starting with an apostrophe may lose it in a spreadsheet; use double quotes for quoted speech.
- **Several games in one file:** the answer-cue figures are given for each game (topic and module) as well as for the whole file, and the milestone exams get their own length check.
- **Other checks:** select-all questions that almost always have the same number of correct options, short answers that the typo rule would confuse with another term, and a results link that is a Google Form, Sheet or test (`/dev`) link instead of the collector's Web app URL (a red item).
- **Results are sent to:** the last line of the check says exactly where results will go (the collector, an email address, or nowhere) and whether names are required. Read it before sharing the file.

Load your file yourself before class and fix the red items. If you work from the repository, `npm run validate-tsv -- my_questions.tsv` runs the same checks on the command line.



The start page



File check after loading the demo file

Generating questions with Claude (optional)

Writing a full question file takes time. The **LAB question-writer** is a skill for Claude (Anthropic's AI assistant) that does the heavy lifting with you:

1. It asks about your course, level, learning objectives and session length, and reads any notes or slides you share.
2. It proposes a board layout (4 themes × 2 subthemes, mapped to your objectives) for your approval.
3. It writes the questions, with distractors based on common misconceptions and explanations that address them. It can also draw simple figures.
4. It checks the file with the same rules as the game and hands you a ready-to-load `.tsv`.

To use it:

- **In Claude (claude.ai or the desktop app):** download the skill zip and add it in Claude's skills settings (Customize → Skills). When I update the skill, download the zip again and replace the old one.
- **In Claude Code:** install it as a plugin. Type `/plugin marketplace add hghezzi/Learn-Around-the-Board`, then `/plugin install lab-question-writer@lab`. To get later versions, run `/plugin marketplace update lab` (or turn on auto-update for the marketplace in `/plugin`). If you work in the project repository itself, the skill also loads by itself.

Then ask, for example: *"Make a Learn Around the Board game reviewing enzyme kinetics for second-year biochemistry."* Always review generated questions before using them in class; you know what your students were actually taught.

Before you share any file with Claude, think about privacy. The skill runs in your own Claude conversation, so everything you upload or paste (notes, slides, spreadsheets, past results) is read by Claude and handled by Anthropic under your Claude account's terms and privacy settings. Learn Around the Board never receives it. So:

- **Remove student information first:** names, student numbers, emails, grades, accommodations and any other personal or sensitive data. The skill keeps such data out of the question file and warns you if it finds some, but by then Claude has already read it.
- **Share only what you're allowed to share:** no live exams, licensed test banks, unpublished research or other confidential material.
- **Use your institution's Claude account if it has one** (Claude for Education or Enterprise), and follow its rules on what may be uploaded. On a personal account, check your privacy settings, including whether your chats may be used to improve Claude, before you upload course materials.

The skill follows your materials: it writes to your course's notation, terms and worked examples, points out errors it finds in your notes before writing, and lists by row which facts go beyond what you gave it so you can check them. It computes every number and runs every code snippet before putting them in a question. It also follows a few ground rules:

- **Student data:** it keeps names, grades and other personal data out of the file, and uses class results only as class-wide misconceptions.
- **Your materials are data, not instructions:** it reports hidden instructions in uploaded notes instead of following them, and it only sends results where you yourself said.
- **Copyright and exams:** it writes original questions rather than copying a publisher's test bank, and won't put upcoming exam questions in a game that shows every answer.
- **Clinical and sensitive topics:** it asks you to check doses against your institution's references and follows safe-messaging practice for topics such as suicide.

I tested these rules with simulated instructors across eleven very different requests (with materials and without, in Spanish, editing an existing file, impossible requests, a roster pasted by mistake, hidden instructions in notes), with independent reviewers checking safety, teaching quality, fidelity to the materials and fit with the game.

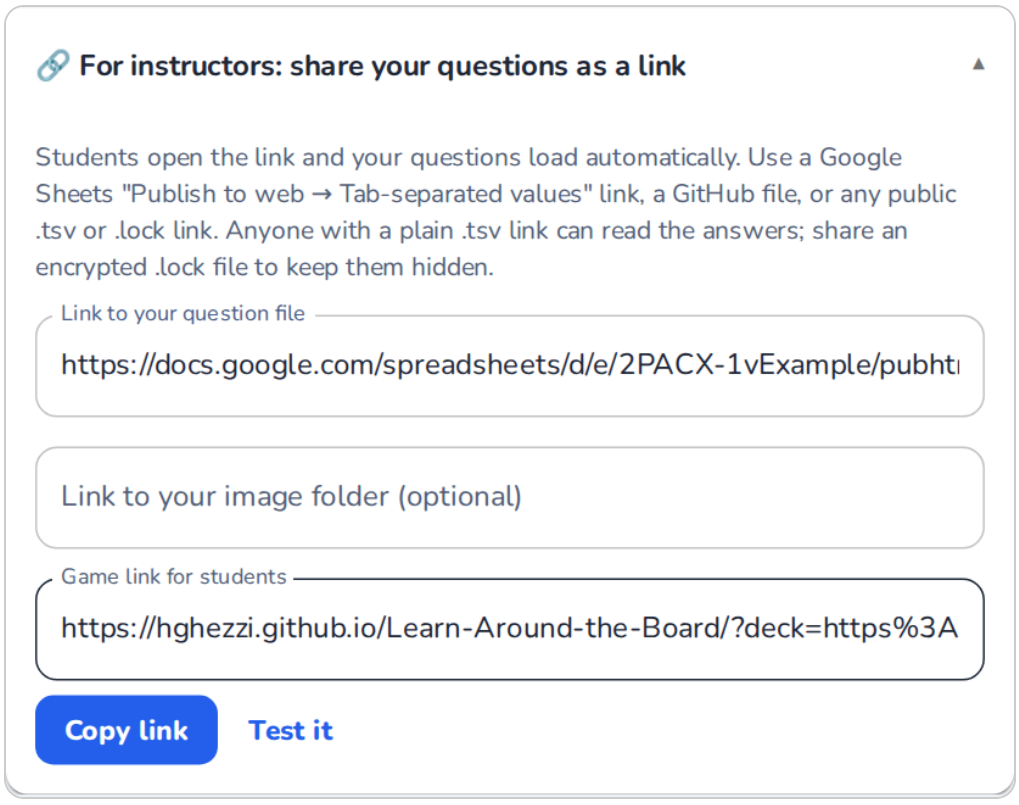
Optional: encrypting your questions file

Since the question file you distribute to students also contains the answers, I created an encryptor tool. Type a class password, choose your `.tsv` file and click **Download .lock File**; the encryption happens in your browser and the file is never uploaded. Share the `.lock` file together with the password; when students open it, the game asks for the class password. Check the plain `.tsv` before encrypting it, and keep it: a `.lock` file can't be opened without its password. Encryption keeps casual eyes off the answer key; it is not meant as strong security, since every student knows the password.

Sharing the game with students

Students can load your questions in three ways:

- **A game link (easiest):** on the start page, open **For instructors: share your questions as a link**, paste a link to your question file into **Link to your question file** and click **Copy link**. Use **Test it** to open the link yourself first. Students click it and your questions load straight away, so you can post it on your course page or put it on a slide.



The link builder on the start page turns a link to your question file into a game link

- **Uploading the file:** students download your `.tsv` or `.lock` file and click **Upload** on the start page.
- **The built-in examples:** `?deck=demo` and `?deck=stats` at the end of the game's address open the demo and the statistics example.

Where can the file live?

- **Google Sheets:** keep your questions in a Sheet, then choose **File → Share → Publish to web**, pick the tab with your questions, choose **Tab-separated values (.tsv)** and click **Publish**. Paste the link it gives you into the link builder. Edits to the Sheet reach new games within about 5 minutes. A normal share or edit link doesn't work: the Sheet must be published.
- **GitHub:** paste the link of the file's page; the game turns it into a link to the raw file.
- **Any public web link** to a `.tsv` or `.lock` file, for example on your own website. Google Drive file links don't work, because Drive shows a preview page instead of the file.

For images, put them in a public folder (for example a GitHub folder) and paste its link in **Link to your image folder (optional)**; the game then looks for each `imageFile` there.

Anyone with a link to a plain `.tsv` can read the answers. To keep them hidden, link to an encrypted `.lock` file instead; students then type the class password when the game opens.

Game mechanics

Once the file is loaded and the game begins, Learn Around the Board operates on a blend of resource management, academic trivia and strategic risk-taking. Here is how the game plays out. The Student Guide explains the same rules to students in a few minutes.

The objective

Players compete to achieve one of two victory conditions:

1. **Last player standing:** be the last player that has not been eliminated by bankruptcy.
2. **Highest net worth:** when the session timer runs out, or when you ask players to click **End game**, the player with the highest net worth (cash plus what it paid for the

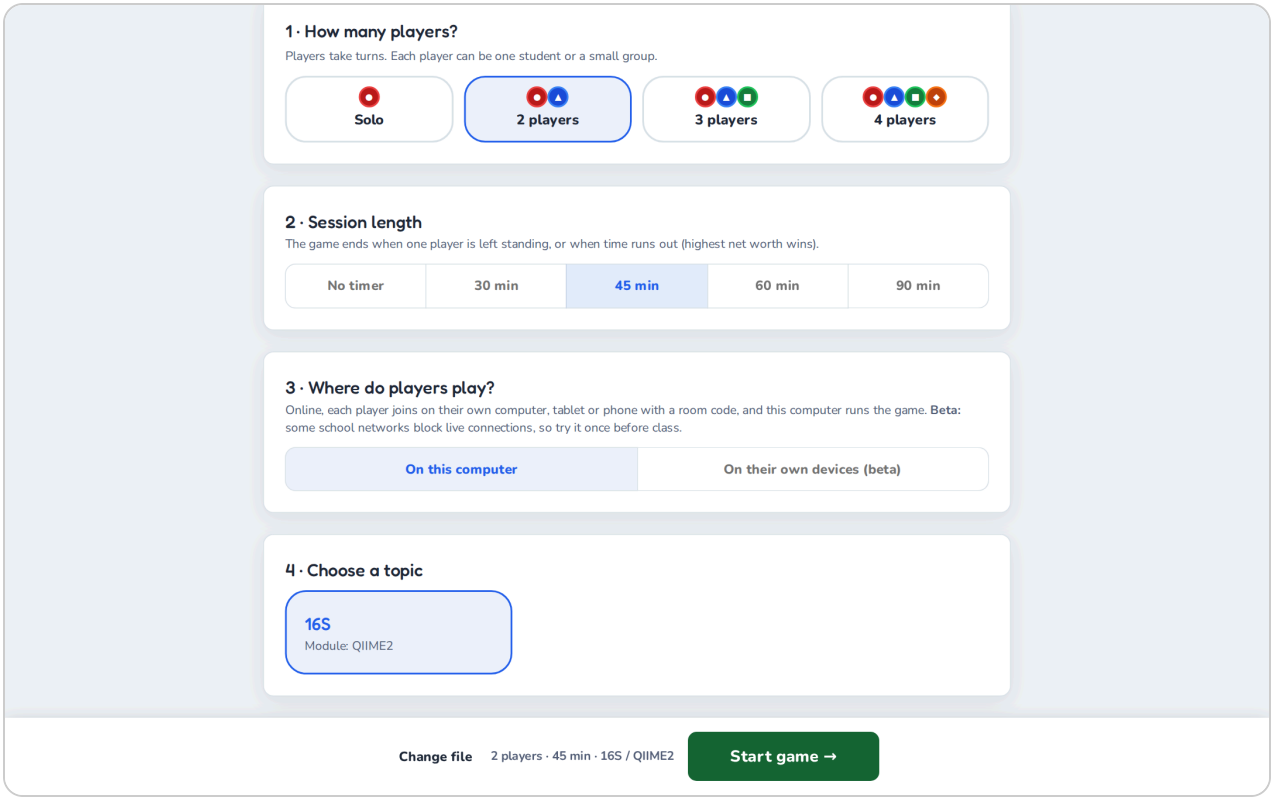
properties and upgrades it still owns) wins.

Milestones don't win the game on their own. They earn Chaos Tokens and add to a player's net worth, so mastery still pays off.

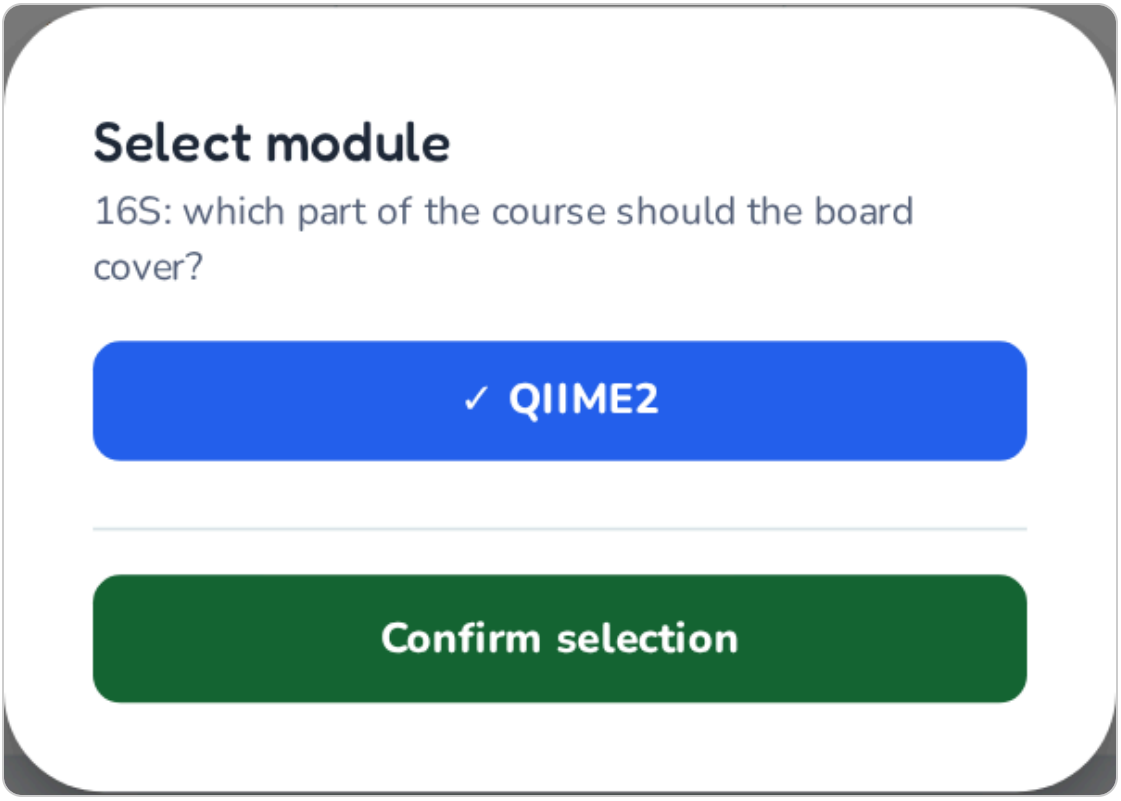
Game setup

At each computer, one student loads the question file, or opens your game link, and adds the optional images, then clicks **Continue to game setup →** and picks:

- **How many players?** *Solo* or 2, 3 or 4 players sharing the computer (each player can be one student or a small group);
- **Session length:** *No timer*, or 30, 45, 60 or 90 minutes;
- **Where do players play?** *On this computer* (players take turns at one screen), or *On their own devices (beta)*: see Online play (beta);
- **Choose a topic**, then the module under **Select module**.

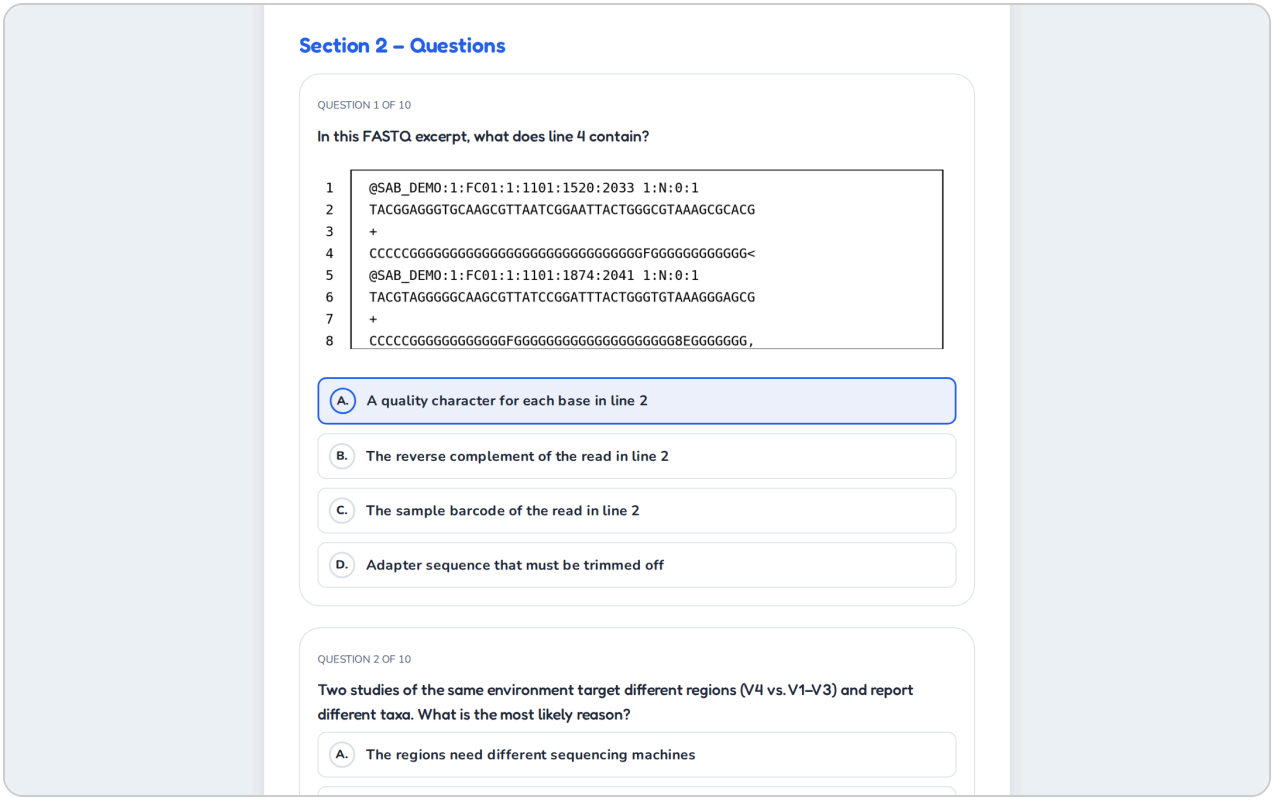


Game setup: number of players, session length and topic



Choosing a module

After **Start game →**, each player completes a brief pre-game survey: the confidence sliders and 10 questions. Players take turns on the same screen until everyone is done, then the board opens. The player with the best pre-game survey score (unknown to players) starts; ties are broken at random.



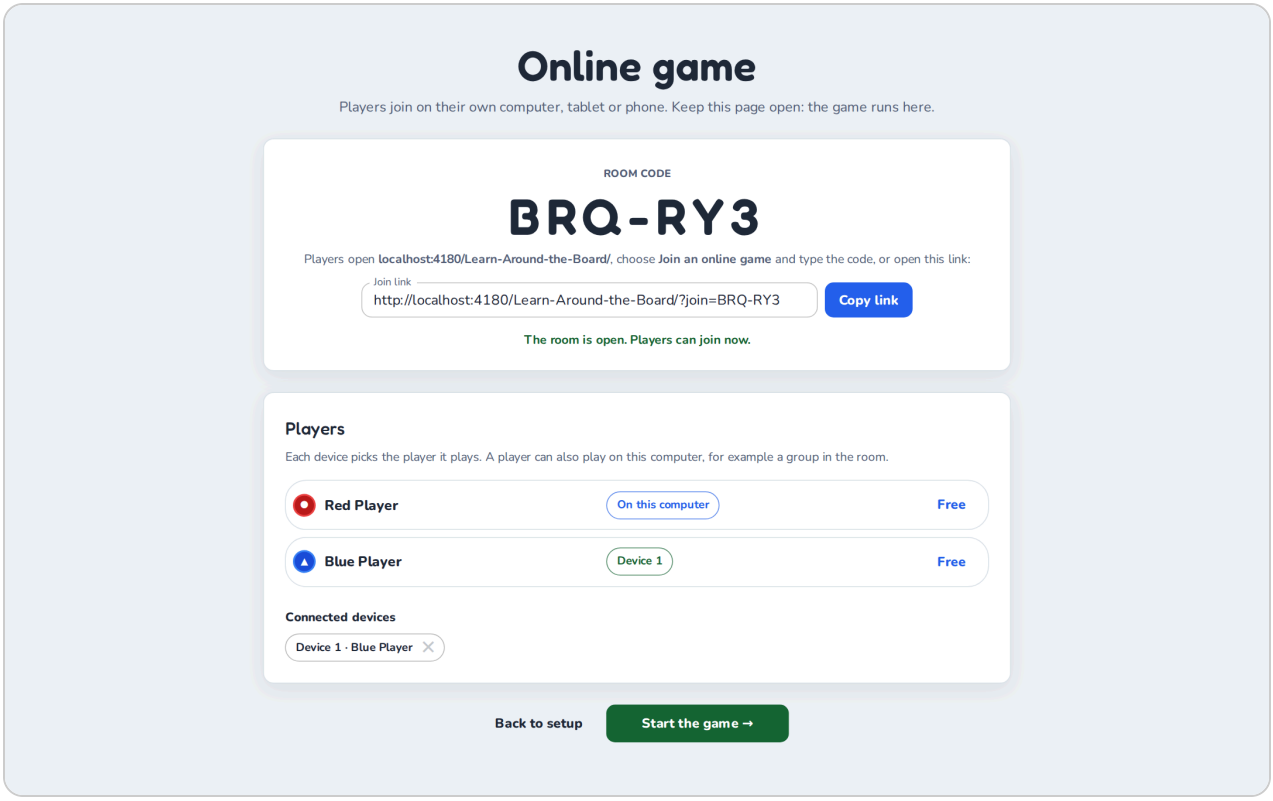
Pre-game survey

Online play (beta)

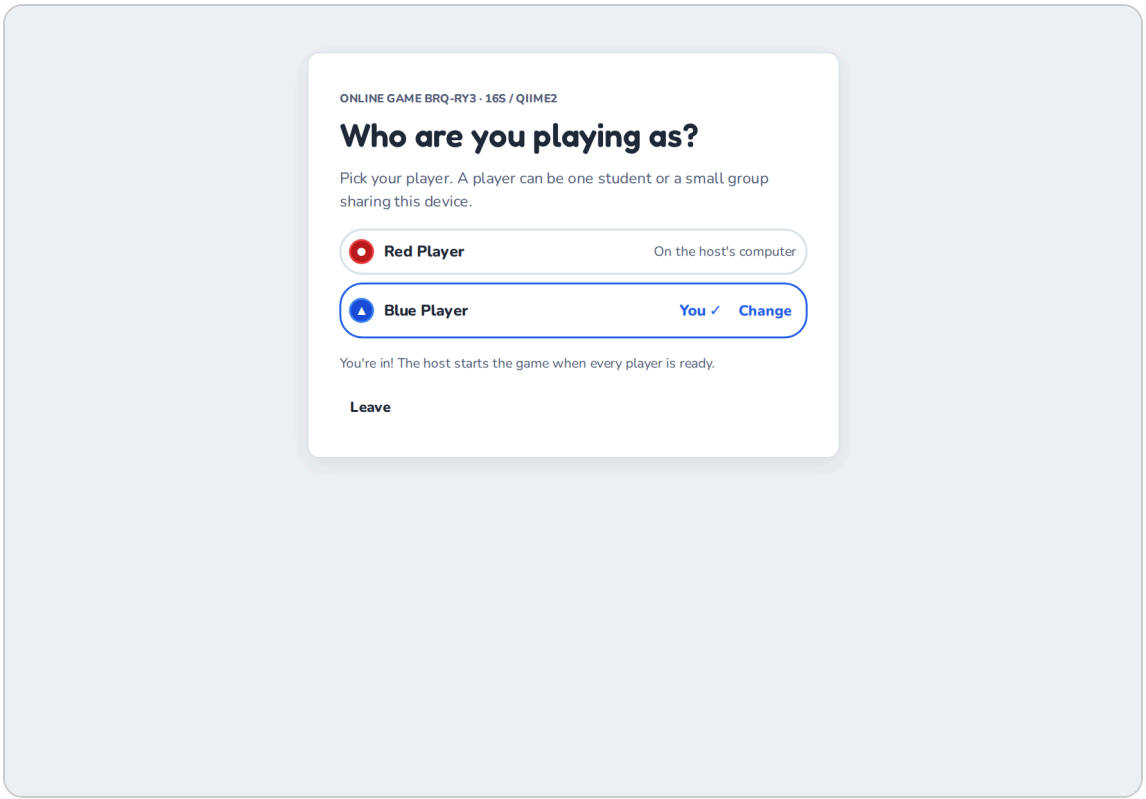
Online, every player can play from their own computer, tablet or phone, in the same room or from home. One computer, usually yours, **hosts** the game: it runs the board, and the other devices show what it shows and send the buttons their players press. I made this for remote classes, and for big rooms where groups can't crowd around one screen.

Hosting a game

1. Load your question file as usual and continue to the setup. Choose the number of players, then **Where do players play? → On their own devices (beta)**, a topic, and **Open the online room →**.
2. The lobby shows a **room code** (for example *V7B-5NB*) and a **join link**. Put the code on the projector or post the link in your course chat.
3. Players open the game, choose **Join an online game** on the start page and type the code (or open the link), then pick their player (*Play as Red Player*, …). A group can share one device, and a player can also play on the host computer: click **Play on this computer** next to it.
4. When every player has a device (or "On this computer"), click **Start the game →**.



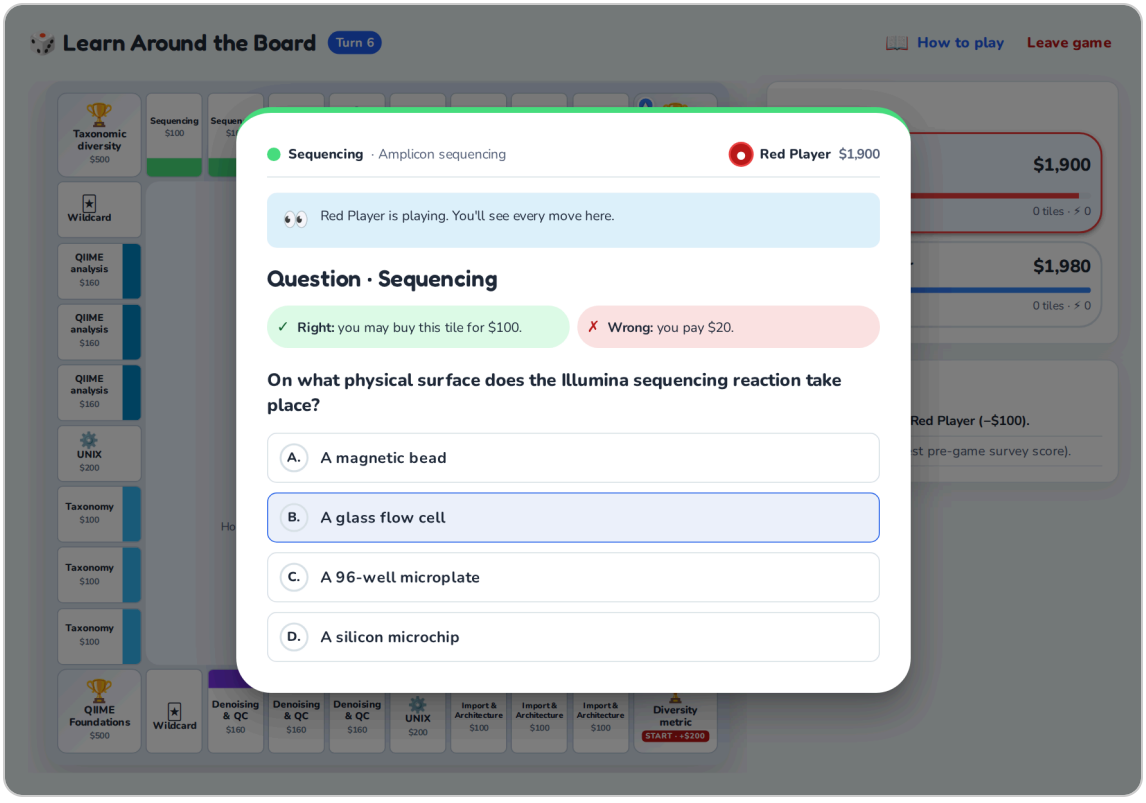
The host's lobby: the room code, the join link and who plays which player



A player's device: picking a player

During the game

- Each device answers its own **pre- and post-game surveys** at the same time, so nobody waits for the computer to be passed around. If a device can't finish (a flat battery, say), click **Answer on this computer** on the host's waiting screen.
- On their turn, players roll, answer, buy, upgrade and use chaos tokens on their own device. Everyone else sees the same board and dialogs, read-only, with a note saying who is playing.
- The host can always act for any player, for example when a device drops out. **End game** and the final standings are on the host only.
- If a device refreshes or loses the connection, it reconnects by itself and gets its player back. If the host computer refreshes, choose **Resume** on its start page: the same room reopens and the devices reconnect.
- At the end, players can type their names or student IDs on their own device and press **Send names to the host**; they appear on the host's results screen, and the host sends or downloads the one results file for the whole game, as usual.



A player's device while someone else answers

Good to know

- **Keep the host page open** for the whole game: the game runs there. Turn off the host computer's sleep mode.

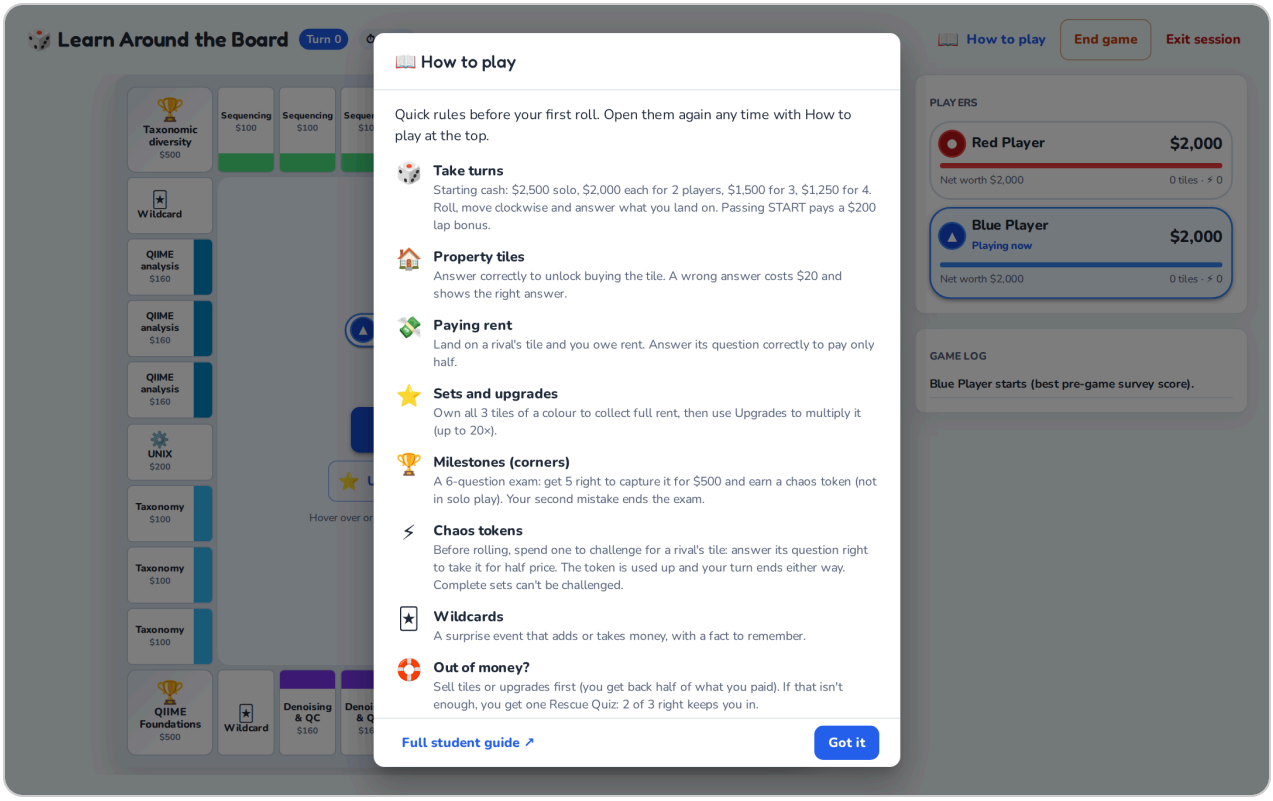
- **Try it once on your own network before class.** Devices find each other through a free public connection service (PeerJS) and then connect directly, encrypted. Some school and company networks block these connections. If a device can't join, a phone hotspot usually works; otherwise, play on one computer, or share the host's screen on a video call.
- **Images:** devices see hosted images, web links in the `imageFile` column and the image folder of a game link, but not images uploaded on the host computer.
- Devices never receive the answers before a question is answered, so a student can't read them from the network traffic.
- So far it has been tested in Chrome on computers. Try the browsers, tablets and phones your students will use before relying on it in class.

The core turn sequence

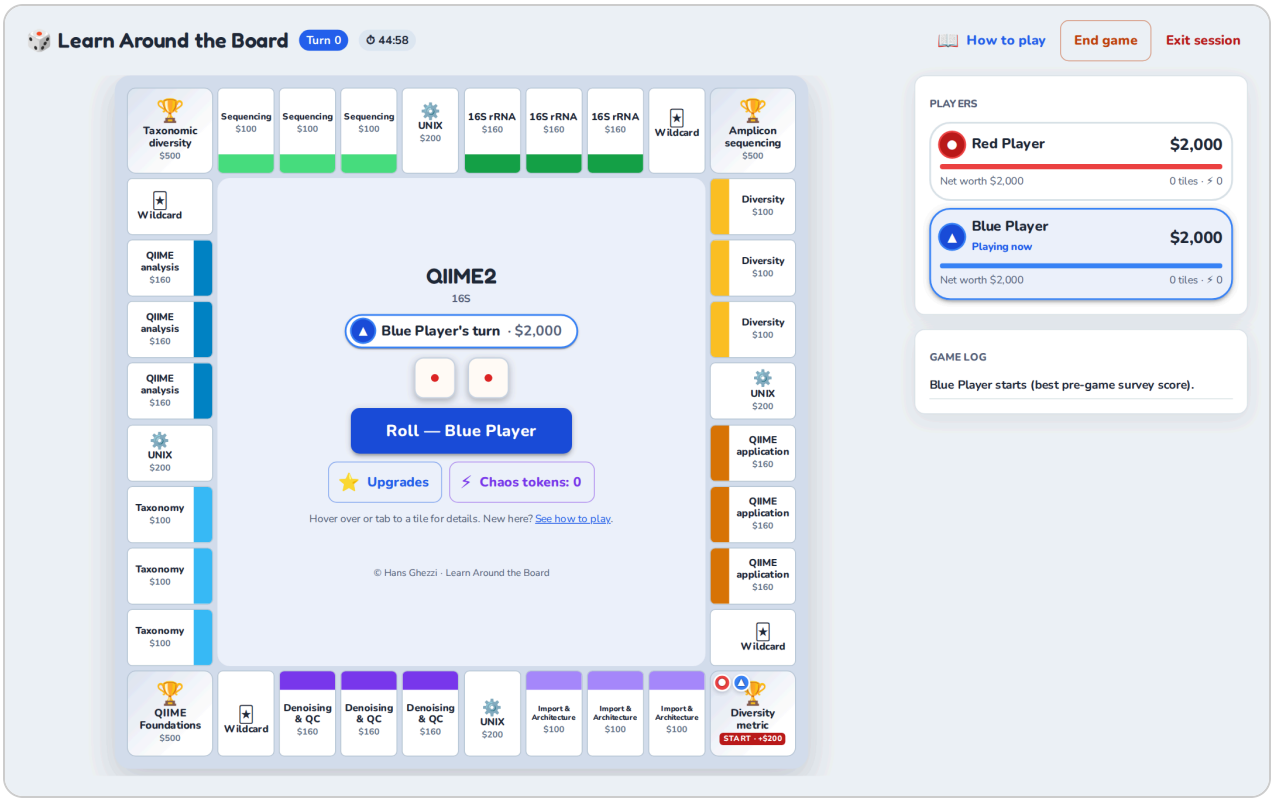
Each player’s turn follows a fast-paced loop:

1. **Roll:** the active player rolls the dice in the centre of the board.
2. **Move:** their pawn advances around the board automatically. Passing START earns a $200 lap bonus.
3. **Encounter:** the player interacts with the tile they land on.

A new game opens with a short **How to play** pop-up of the quick rules, and players can reopen it at any time with **How to play** at the top of the board (the start page has a **Quick rules** link too).



The quick rules that open at the start of a game

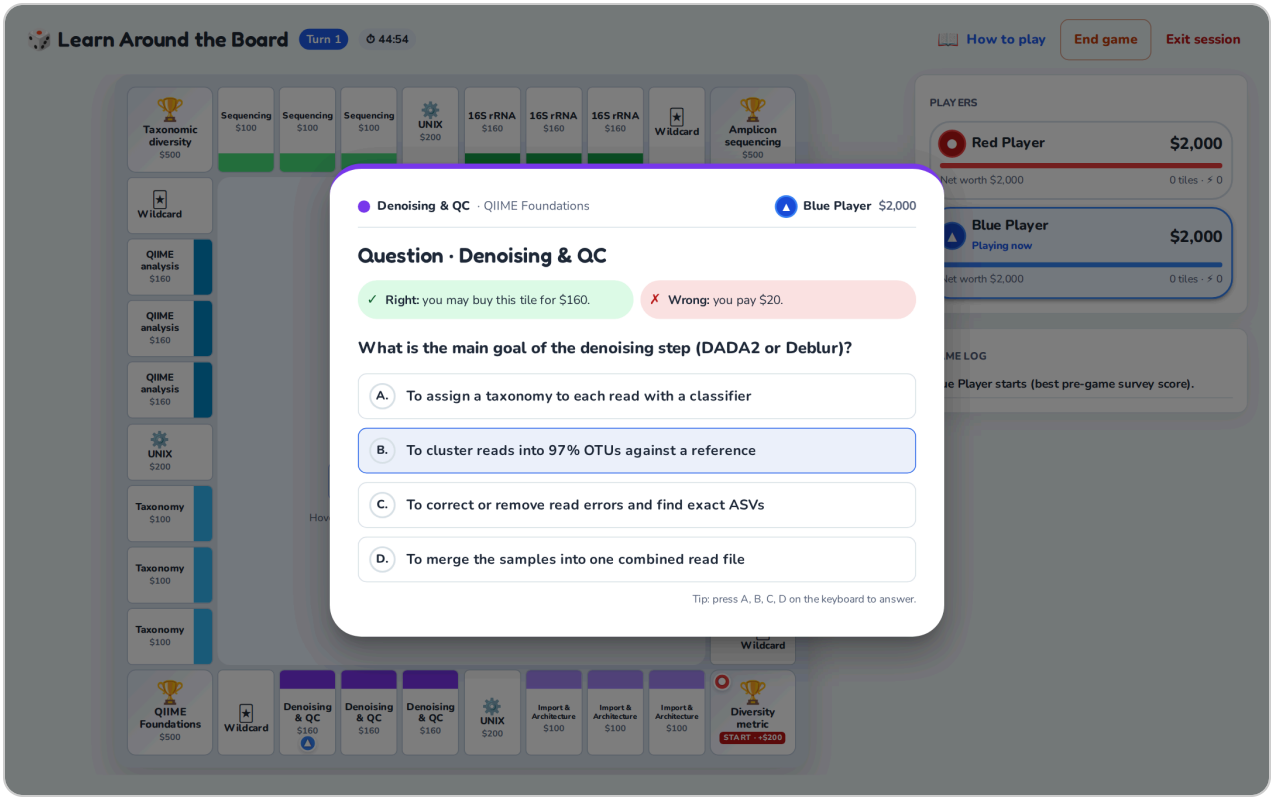


The board. The centre shows whose turn it is, the dice and tile details; the panel on the right shows each player's cash and net worth

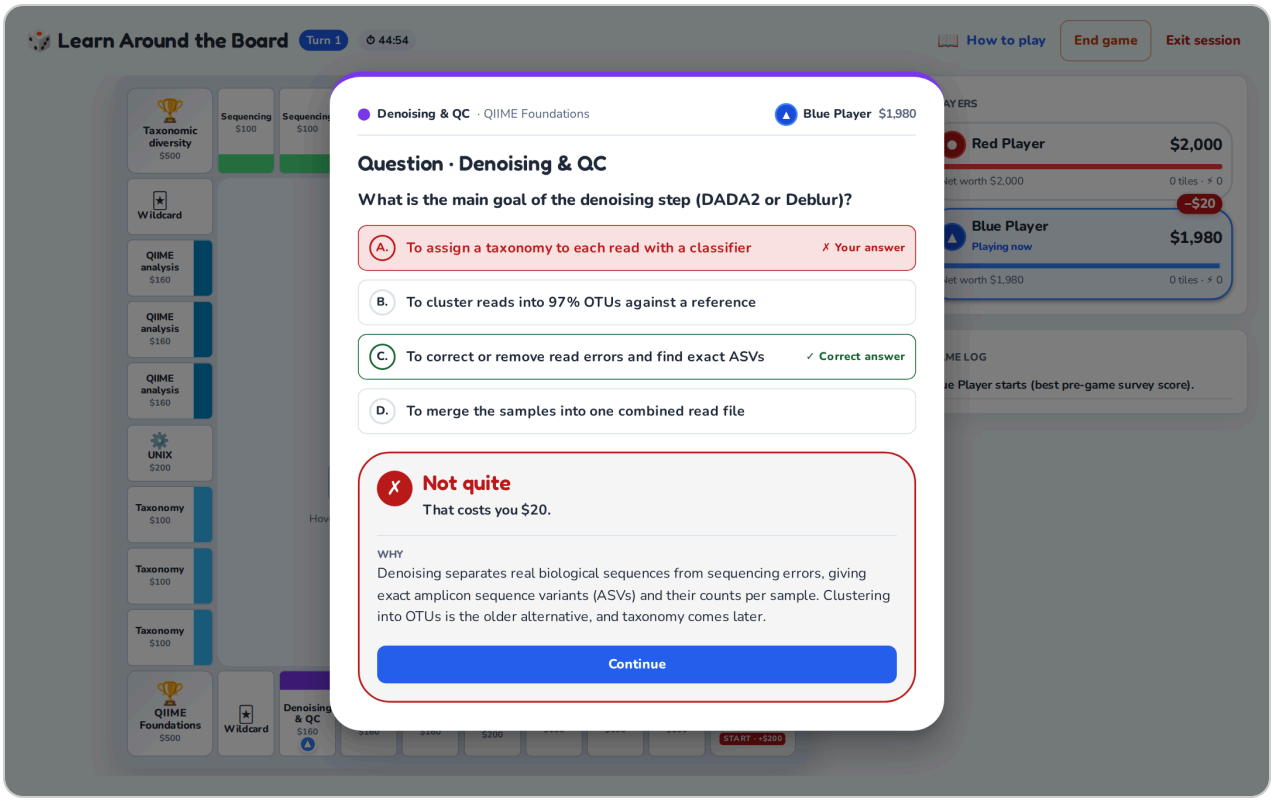
Tile encounters

The heart of the game lies in what happens when a player lands on a tile.

- **Unowned tiles (properties):** the player may try to buy the tile. They are shown one of your questions. Questions nobody has seen yet come first, and a question a player got wrong comes back about six turns later, so missed ideas get a second, spaced retrieval.
  - *Correct answer:* they may pay the tile's price and take ownership, or skip.
  - *Incorrect answer:* they pay a $20 fine, the tile stays unowned, and the correct answer and explanation are shown.
- **Rival tiles (rent defense):** landing on a rival's tile means paying rent, but the player first answers a question.
  - *Correct answer:* rent is reduced by 50%.
  - *Incorrect answer:* the player pays full rent.
- **Milestones (corner exams):** capturing a corner requires more than money; it requires mastery. A player needs $500 to attempt the exam and must answer 5 out of 6 questions on that side's theme; a second mistake ends the exam. A player that passes pays $500, captures the corner and earns a Chaos Token; a player that fails pays nothing. Landing on a rival's milestone means a $250 fee, or an expert challenge (the same 6-question exam) that halves it.
- **Core tiles:** tiles with questions from your `core` rows. They are bought ($200) and defended like properties, with a rent of $50 for each core tile the owner holds ($50 with one, $100 with two, up to $200 with all four).
- **Wildcard tiles:** draw a random card from your `mishap` rows that pays or costs the amount written on it (e.g. "Scholarship awarded! +$200" or "Contamination! -$100"), with a fun fact.



A question. Options are shuffled each time; other formats show checkboxes, a number box, a text box or a list to reorder



Feedback after an answer, with the correct answer and your explanation

Advanced mechanics: upgrades and chaos

To keep engagement high in the later stages of the session, the engine includes two advanced mechanics. Players use both from the buttons in the centre of the board during their turn, before rolling.

- **Upgrades:** if a player owns all the tiles in a subtheme (e.g. all three "Denoising & QC" tiles), they can use the **Upgrades** button to upgrade those tiles evenly, up to four stars. This drastically increases the rent charged to rivals who land there.
- **Chaos Tokens:** capturing a milestone awards a Chaos Token. A token can be spent on a Chaos Challenge (the **Chaos tokens** button) against a rival's property: answer a question from that property's own subtheme correctly to take it for half its price (paid to its owner), or pay a small fine. Either way the token is used up and the player's turn ends without a roll. A complete set (all three tiles of a subtheme, upgraded or not) is protected and can't be challenged, and the Challenge button stays off until the player can afford the price. Once all four milestones have been captured, Chaos Tokens can be bought for $500.

Bankruptcy and the Rescue Quiz

If a player's balance goes negative:

1. **Liquidation:** the player must sell properties or remove upgrades until it is out of debt. Each sale returns half of what the player paid for it.
2. **Rescue Quiz (once per game):** if selling everything still can't cover the debt, the player takes a 3-question Rescue Quiz drawn from the whole board. With 2 or more correct answers, the debt is cleared and the player receives $500 to keep playing.
3. **Elimination:** a player who fails the Rescue Quiz, or goes bankrupt a second time, is eliminated, and their properties return to the bank. Those students still take the post-game survey.

Playing solo

A single player (one student, or a small group sharing one player) has no rivals to pay rent or to challenge, so solo play changes three things:

- **Your own tiles pay.** Landing on a tile you own asks one of its questions. A right answer collects its rent from the bank, the same amount a rival would pay (more with a

full set, upgrades and more core tiles; $250 on your own milestone). A wrong answer costs $20. Buying and upgrading become investments that pay off only if you know the material, and every roll keeps asking questions.
- **A goal to beat the clock.** The board shows a net-worth goal: $3,300 in a 30-minute session, $4,000 in 45, $5,000 in 60 (or with no timer) and $9,000 in 90. I sized them with a simulation so that most students answering about 80% correctly reach the goal, about a third at 65%, and few at 50%. Capturing all four milestones is a bonus.
- **No chaos tokens,** since there is nobody to challenge.

The end screen shows whether the goal was reached, the share of questions answered correctly, the milestones captured and the best result so far on that computer (kept only in that browser).

Playing against the bot

On the setup screen, a solo player can choose **Play against the bot** instead of **Reach a goal**, and pick **Easy**, **Medium** or **Hard**. The bot then plays the normal two-player game: it buys, charges rent, upgrades, takes milestone exams and uses chaos tokens, with $2,000 each to start. The student always goes first.

- **The bot answers at its level:** about 50% right on Easy, 70% on Medium and 85% on Hard. I tuned the levels with a simulation of the real rules: a student who answers about as well as the bot wins about half the time, and every extra 10% of right answers adds roughly 15 points to the student's chance. A student at 70% wins about 3 games in 4 against Easy, half against Medium and 1 in 3 against Hard.
- **Its turns are worked examples.** The student sees each of the bot's questions, its answer and the explanation, but can't answer or press anything; a turn takes about 5–15 seconds, longer for a milestone exam. **Skip ahead** speeds up the rest of that turn.
- **Only the student's answers count.** The bot's answers are never written to the results file, never count in accuracy or surveys, and its questions can still come up for the student later. The bot's moves appear as `TRANSACTION` rows, and its final result is a `GAME_RESULT` row with `bot` set to its level.
- The end screen shows who won and the student's record against that level on that computer (wins and games, kept only in that browser).
- The bot plays only on one computer; it isn't available in online games.

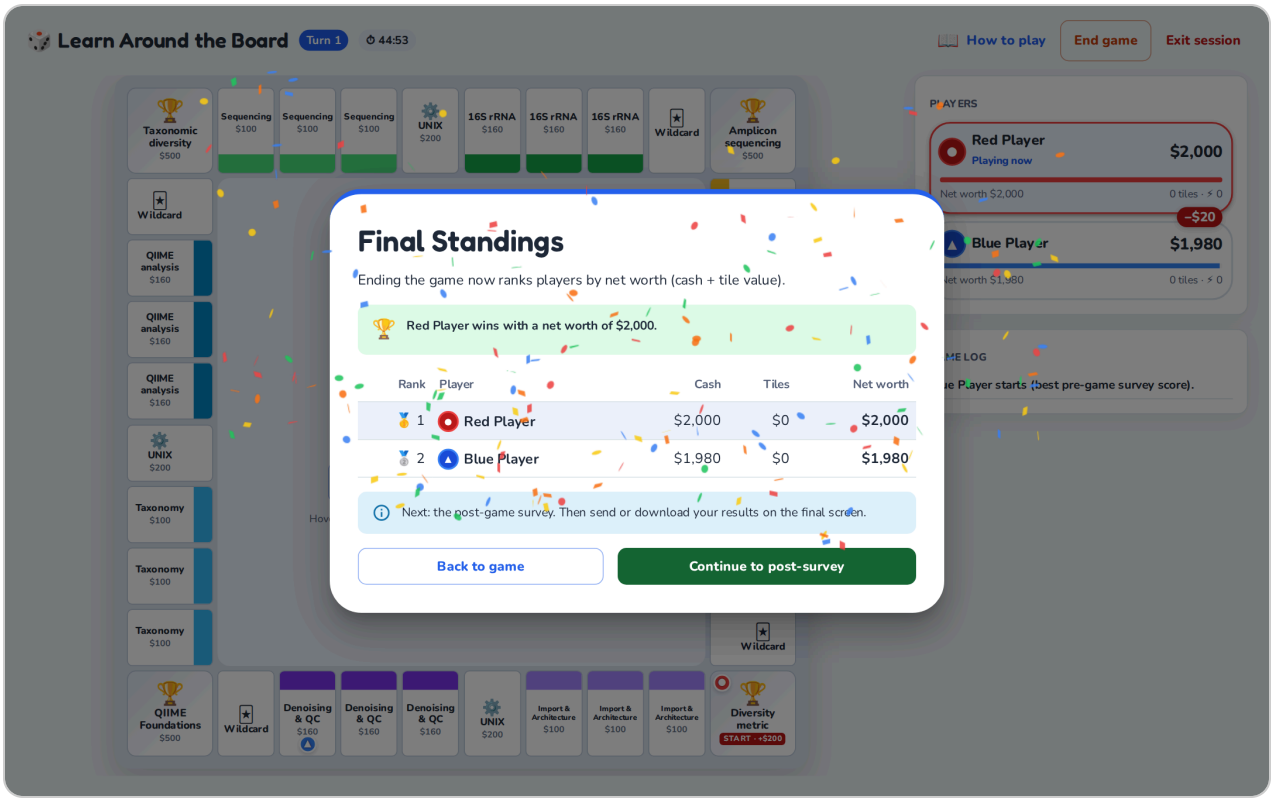
Numbers at a glance

	Amount
Starting cash (each player)	$2,500 solo · $2,000 with 2 players · $1,500 with 3 · $1,250 with 4. Fewer players get more turns each and need more cash; these amounts, chosen with a simulation of 3,000 games per setting, keep the pressure on cash similar for 2–4 players
Lap bonus (passing START)	+$200
Tile prices	$100 (first subtheme of a side) · $160 (second subtheme) · $200 (core) · $500 (milestone)
Wrong answer when trying to buy	−$20
Property rent	50% of the price ($50 or $80); half that until the owner has the whole group; ×3, ×6, ×10, ×20 with 1–4 stars
Core rent · rival milestone fee	$50 per core tile the owner holds (up to $200) · $250
Correct rent-defense answer or passed expert challenge	pays half
Upgrade cost (paid once for the whole group)	the tile price for each of stars 1–3, twice the price for star 4
Chaos steal · failed challenge · buying a token	half the tile’s price · half its base rent · $500 (complete sets can’t be challenged; the token is spent either way and the turn ends)
Liquidation	tiles and upgrades sell for half of what was paid for them
Rescue Quiz	2 of 3 correct: debt cleared and +$500 (once per player)
Solo play	right answer on your own tile: the bank pays its rent · wrong: −$20 · goal: $3,300 / $4,000 / $5,000 / $9,000 net worth in 30 / 45 / 60 (or no timer) / 90 minutes

	Amount
Net worth	cash + what the player paid for the tiles and upgrades it still owns (an upgrade counts once, as it was charged; a tile taken with chaos counts at the half price paid)

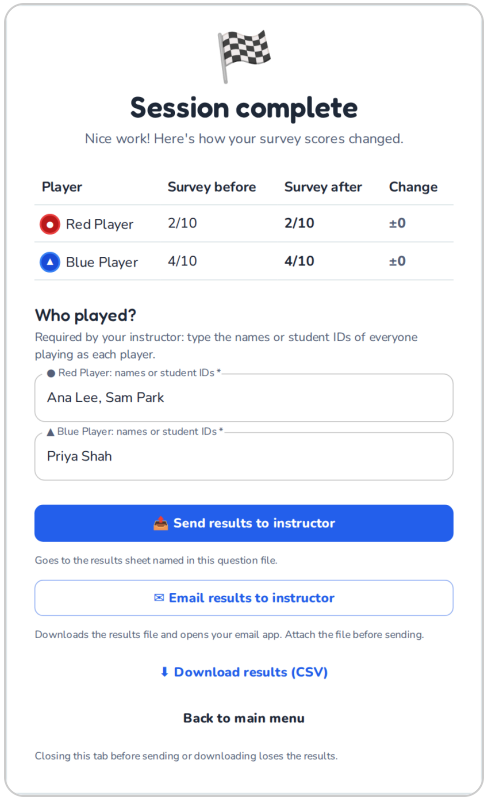
Endgame dynamics

The game ends when only one player is left standing, when the session timer runs out (the timer turns orange in the last 5 minutes), or when a player clicks **End game** and confirms with **Yes, end the game**. A **Final Standings** screen ranks the players by net worth and names the winner. After **End game**, players can still go **Back to game** if they clicked it by mistake; **Continue to post-survey** moves on.



Final standings

Next, every player completes the post-game survey: the same confidence sliders and questions as before the game. The end screen (**Session complete**) then shows each player's survey scores before and after the game, and students hand in their results.



The results screen when the question file sets a results Sheet and an email address. Without config rows, only Download appears

Collecting results

On the end screen, students type the names or student IDs of everyone playing as each player, then hand in the results in one of three ways. You choose which ones appear with config rows in your question file (see Config rows):

Option	What students do	What you set up
Google Sheet (recommended)	click **Send results to instructor**	a results Sheet with the collector script, once per course (about 5 minutes), and a `results_url` config row
Email	click **Email results to instructor**: the results file downloads and their email app opens a draft addressed to you; they attach the file and send it	an `instructor_email` config row
Download (always available)	click **Download results (CSV)** and submit the file where you ask, for example on your course page	nothing

When a results link or email address is set, names are required before **Send**, **Email** or **Download** (add an `ask_names` row set to `no` to make them optional). With download-only results, add an `ask_names` row set to `yes` if you want names in every file you collect. The game can't insist on student numbers rather than names; say which you want.

Setting up the Google Sheet collector

1. Create a new Google Sheet, for example "LAB results – BIOL 301".
2. Choose **Extensions → Apps Script**. Delete the sample code, paste the collector script and click **Save**.
3. Choose **Deploy → New deployment**, select the type **Web app**, set *Execute as* to **Me** and *Who has access* to **Anyone**, then click **Deploy** and authorize the script with your Google account.
4. Copy the **Web app URL** (it ends in `/exec`) and add it to your question file as a config row: `id` = `results_url`, `type` = `config`, `question` = the URL.
5. Optionally, add `instructor_email` and `course` rows as well.
6. Test it: load your file, play a quick solo game and click **Send results to instructor**. Two tabs appear in your Sheet: **Summary** (one row per player, with names, survey scores and rank) and **Details** (every answer and transaction).

If you edit the script later, use **Deploy → Manage deployments → Edit → Version: New version**, so the URL stays the same. Every submission carries a `sessionId`, and a game sent twice is recorded only once.

For safety, the game sends results only to a Google Apps Script Web app; it ignores any other address, so a question file passed around by someone else can't send your students' names and answers to their own server. The start page says when a file sends results (and which course it names), and a file opened from a link shows which site it came from. Share your files and links only through your course page.

Results are what the students' browsers send, so a determined student could alter them: use them for participation and to see what the class understood, not as proof of a score.

Who has access: Anyone lets anyone with the link send data to your Sheet, but only you can read it, and the collector accepts only game submissions. If your institution doesn't allow Google services for student data, use the email or download option instead, and ask students to type student numbers or initials rather than full names if your rules require it.

What the results contain

Learn Around the Board collects feedback on student performance in a Comma-Separated Values (CSV) file named after the topic, module, date and time, for example `lab_results_16s_qiime2_2026-10-06_1430.csv`; the **Details** tab of the results Sheet holds the same rows. It tracks every answer and transaction during the game, plus the pre- and post-game survey results. Instructors can open it in Excel or Google Sheets, or analyse it with R or Python, to see which questions were missed most often and how confidence and knowledge changed.

Each row has an `eventType` (game events) or a `phase` (surveys):

Row	What it records
`TEAM_INFO`	One row per player: the names or IDs typed on the end screen, survey scores before and after (`preScore`, `postScore`), `rank` and `netWorth`
`phase` = `pre` / `post`, `section` = `confidence`	Each player's slider value (`response`) for each confidence statement
`phase` = `pre` / `post`, `section` = `quiz`	Each survey question: `questionId`, `selectedOption`, `selectedIndex` (for multiple choice, the option number in your file, 1–4, like `correctIndex`), `correctAnswer`, `correct`
`PROPERTY_Q`, `RENT_Q`, `OWN_TILE_Q`, `MILESTONE_Q`, `CHAOS_Q`, `GRANT_Q`	Every in-game question (`OWN_TILE_Q`: a solo player's own tile): player (`playerName`), `questionId`, `format`, `response`, `correctAnswer`, `correct`, tile
`TRANSACTION`	Every money change, with the reason (`action`), `amount`, `moneyBefore`, `moneyAfter` and `timestamp`
`ELIMINATED`	A player leaving the game, and why
`SOLO_GOAL`	Solo only: the `turn` the net-worth goal was reached, and the `netWorth` at that moment (bank rent appears as `TRANSACTION` rows with `action` = `SOLO_RENT`)
`GAME_RESULT`	Final `rank`, `cash`, property value (`assets`) and `netWorth` for each player, and how the game ended (`endReason`). Against the bot, the bot's row has `bot` set to `easy`, `medium` or `hard`

(`GRANT_Q` rows are Rescue Quiz questions. Some codes, such as `GRANT_Q`, `EMERGENCY_GRANT` and `LAB_MISHAP` for wildcards, keep their original names so that older results files stay comparable.)

Three quick analyses

- **The hardest questions:** filter the rows with a `questionId` and make a pivot table of `correct` by `questionId`. The questions most often missed are your next lecture's warm-up.
- **Learning gain:** compare `preScore` and `postScore` on the `TEAM_INFO` rows, or the `pre` and `post` quiz rows question by question.

- **Calibration:** compare the confidence sliders with the survey scores, before and after.

Grading. I recommend grading participation (a complete results file, or a player appearing in your Sheet) or the growth from pre- to post-game survey, rather than the final rank: dice decide part of every game, and low stakes are what make students willing to take risks.

The game saves the session in the browser as it goes, so an accidental refresh offers **Resume your game?** on the start page. Students should still send or download their results before closing the tab: the saved copy stays on that computer until then, and expires after 12 hours. Sending or downloading the results deletes it, so the next student on a shared computer doesn't see the previous game.

Designing a session based on ‘Learn Around the Board’

Designing a successful Learn Around the Board session requires intentionality, ensuring that the activity and the questions align with the learning objectives of the class or course. Here is a framework to design your session and enhance its pedagogical impact.

Backward design: aligning with Learning Outcomes (LOs)

Before writing a single question, start with your Learning Outcomes. What exactly should students be able to do by the end of this session? Once your LOs are clearly defined, map them directly to the game’s mechanics:

- **Foundational knowledge (properties):** use standard property tiles to test basic recall and terminology. If an LO is "Define the function of 16S rRNA", it belongs on a property tile.
- **Synthesis and application (milestones):** the four corner milestones represent mastery. Reserve your highest-order thinking questions for these exams. If an LO is "Analyze a pipeline output to troubleshoot denoising errors", that scenario is better suited to a milestone quiz. Ordering questions (put the steps of a workflow in order) work especially well here.
- **Practical realities (wildcards):** use your `mishap` rows to teach practical realities or common pitfalls that do not fit neatly into a question format (e.g. "You forgot to balance

the centrifuge! (-$100)").

Learn Around the Board was originally designed for review-style sessions, so it is naturally better suited to “recall”, “understand” or “apply” learning objectives than to “create” levels.

Crafting the question file

When writing your TSV file, remember that the "incorrect" options are just as important as the correct one.

- **Write plausible distractors:** do not use throwaway joke answers. Every incorrect option should represent a common student misconception. When a student chooses a distractor, it reveals a specific gap in their mental model.
- **Don't let the answer give itself away:** keep every option the same length, detail and grammar as the correct one. A careful author tends to make the correct answer the longest and most qualified option, and students quickly learn to pick it. Cover the question, read only the options, and ask whether you could guess. The file checker measures this for you.
- **Leverage the explanation column:** this is your most powerful teaching tool in the game. Do not just write "Incorrect". Use this space to immediately correct the specific misconception tied to the distractors. Immediate, targeted feedback at the exact moment of failure is what transforms this from a quiz into a learning tool.
- **Write separate survey questions:** make the `survey` questions parallel to the board content (same LOs) but not copies of board questions, so the post-game survey measures learning rather than memory of an exact question.

Adapting the game to any subject

The game’s own words (Wildcard, Upgrades, Rescue Quiz, dollars) are deliberately neutral; everything subject-specific comes from your file. A few examples:

Course	Themes (board sides)	A wildcard
History	Eras, such as "The Bronze Age" or "The Industrial Revolution"	*"The library of Alexandria burns! (-$100)"*
Computer science	Algorithms, data structures, languages	*"You pushed straight to production on a Friday. (-$150)"*
Literature	Authors, movements, genres	*"Your book club finally finished Ulysses! (+$100)"*
Statistics	See the built-in Intro Statistics example	*"Your p-hacking was caught in peer review. (-$100)"*

Structuring your time: the BOPPPS model

To ensure the game serves an academic purpose, I highly recommend structuring your session around the BOPPPS model. For a standard 80-minute block, this translates to:

1. **Bridge-in (5 mins):** hook the students. Explain why you are using a game today (e.g. to synthesize the last 4 weeks of material) and outline the victory conditions.
2. **Outcomes (2 mins):** explicitly state the LOs on the board so students know what they are meant to be learning.
3. **Pre-assessment (10 mins):** have students load the game and complete the mandatory pre-game survey. This establishes a metacognitive baseline for their confidence and current knowledge.
4. **Participatory learning (45 mins):** this is the gameplay phase. Set the session timer to 45 minutes so the game ends on time by itself. Step back and become the Game Master, and let the competitive tension and "hard fun" drive the engagement. Encourage players to read the explanations out loud.
5. **Post-assessment (10 mins):** when the timer ends the game, students check the final standings, complete the post-game survey and send or download their results.
6. **Summary / debrief (8 mins):** do not skip this! Ask the room: "Which side of the board was the hardest to defend?" or "What was the most surprising thing you learned from a mistake?" Harvesting the play into conscious realization is critical.

While this is only an example, you can certainly explore other lesson-design strategies such as the CARD model or 5E lesson planning.

Common challenges

1. **Dark mode:** no longer a problem. The game follows each computer’s light or dark setting and is readable in both. (Older versions hid questions in dark mode.)
2. **Lost progress:** the game saves the session in the browser between turns. After an accidental refresh, the start page offers **Resume your game?**; a refresh in the middle of a turn returns to the start of that turn, and uploaded images have to be selected again. **Exit session** and **Back to main menu** discard the saved game on purpose.
3. **The file won't load:** check that the column names are spelled exactly as above and that the file was exported as tab-separated. The file check on the loading screen explains most problems.
4. **Answer options show `#NAME?`:** the spreadsheet turned text that starts with `-`, `+` or `=` into a formula. See Spreadsheet pitfalls.
5. **Images are missing:** check that filenames match exactly (including capitals) and that students selected all the image files at once. Questions still work without their images.
6. **An encrypted file won't open:** passwords are case-sensitive; share the password exactly as you typed it in the encryptor.
7. **The board looks wrong:** you probably have fewer or more than 4 themes, or more than 2 subthemes per theme, in the chosen module. The file check lists these.
8. **The board is cramped:** the board needs a window at least about 600 pixels wide, so laptops, desktops and tablets in landscape work, and phones don't. Students can zoom out (Ctrl/⌘ and −).
9. **Installing the app:** in Chrome or Edge, click **Install as an app** on the start page (or the install icon in the address bar). On an iPad, use Safari’s **Share → Add to Home Screen**. After the first visit, the game also works offline, although a game link needs the internet to fetch your questions.
10. **A game link won't load:** for Google Sheets, use **File → Share → Publish to web** with *Tab-separated values*, not a normal share link. For other links, check that the file is public, for example by opening the link in a private browser window.
11. **"Send" says it couldn't reach the sheet:** students can use **Email** or **Download** instead. Check that your deployment's *Who has access* is set to **Anyone** and that `results_url` is the Web app URL ending in `/exec`.
12. **Which version will my students get?** Always the current one: the game updates itself on a student’s next visit. Your question file only changes when you change it.
13. **A device can't join an online game:** check the code, and that the host's game page is still open. *"No game with this code is open"* means the room isn't open (or was closed); *"Couldn't connect"* usually means the network blocks live connections: try a

phone hotspot, or play on one computer. If a device says it has a different version, reload that page.

Privacy and student data

The game runs entirely in the students’ browsers. There is no account and no game server. In an online game, devices find each other through the free PeerJS connection service, which sees the devices' internet addresses and the room code but not the game; the game itself then goes directly from device to device, encrypted. Question files, answers and surveys stay on the students’ computers; results leave them only when students click **Send results to instructor** (they go straight to your own Google Sheet) or submit the results file themselves. The autosaved copy of a session also stays in the browser until results are sent or downloaded, and expires after 12 hours. Usage analytics (Google Analytics, with cookies: visits, device type, country) are loaded only if a visitor clicks **Allow analytics** on the start page. If your institution has rules about student data, ask students to choose **No thanks**; the game works the same either way. See the privacy notice.

License and reuse

Learn Around the Board is free for teaching, learning and research, and that won't change. Universities, colleges, schools and other public or charitable institutions can use it, adapt it and even host their own copy, whatever their funding. The code is under the PolyForm Noncommercial License 1.0.0, and this guide, the example question files and the images are under CC BY-NC-SA 4.0: please credit me and share adapted guides or question files under the same terms.

- **Your question files are yours.** A file you write, by hand or with the question-writer skill, belongs to you, and you choose how to share it.
- **Commercial use needs my permission:** selling the game, hosting it for paying customers or building it into a paid product. Get in touch through GitHub.
- If you publish a modified version, give it a different name.

The full terms are in the LICENSE file.

Conclusion

Stepping away from the podium and handing control of the review session over to a board game can feel like a leap of faith. As instructors, we are often trained to deliver content efficiently, and the noisy, debate-filled environment of a gamified workshop is a stark contrast to a quiet lecture hall. However, as you will quickly discover during your first session with Learn Around the Board, that noise is the sound of active, collaborative learning.

By leaning into playful pedagogy, we accomplish much more than simply reviewing material; we fundamentally change the students' relationship with the content. We replace the anxiety of high-stakes testing with the "hard fun" of strategic competition, loss aversion and immediate feedback. The magic circle of the game board creates a space where failure is not only safe, but structurally necessary for growth.

Whether your students are defending the nuances of a microbiome data pipeline, troubleshooting laboratory equipment, or dominating a pop-culture trivia night, the underlying mechanism remains the same: they are engaged, they are thinking critically, and they are harvesting their play into conscious understanding.

Building a great TSV cartridge takes intentional design, and mastering the role of the game master takes practice. But the payoff, watching your students actively conceptualize complex material rather than passively memorizing it, is entirely worth the effort.

Welcome to the board, and have a great session!

What’s new

October 2026 (update 13): steadier online games and a safer results sheet

- **Online play ignores out-of-turn and repeated presses.** Each device's button now counts only on the screen it belongs to: a double tap on a slow connection runs once, and a press that arrives after the game has moved on is ignored with a "try again" note. A device can no longer, by accident or on purpose, buy during a rent question, skip rent or start the Rescue Quiz early. All devices need the latest version (they're told to reload if not).

- **Results sheet script, version 3:** a game sent twice is recorded once, a daily row limit protects the Sheet from floods, and the script's own columns (`receivedAt`, `course`…) can't be overwritten by a submission. To update, paste the new script over the old one and choose **Deploy → Manage deployments → Edit → Version: New version** (the URL stays the same). The old script keeps working.
- **Big question files resume reliably.** The autosave stores the question file once and only the game's progress after each turn, and a notice appears if the browser can't save (storage full).

October 2026 (update 12): safer results and shared computers

- **Results go only to a results Sheet collector.** A `results_url` that isn't a Google Apps Script Web app (`https://script.google.com/macros/s/…/exec`) is now ignored by the game and shown as a red item by the file check. Before, any web address was used, so a question file shared by someone else could send students' names and answers to their own server. The collector links from the setup steps keep working unchanged.
- **Students see where their results go.** The start page names the course a file sends results for, and says which site a file opened from a link came from. The end screen names the course next to **Send results to instructor**.
- **Shared computers:** sending or downloading the results deletes the saved copy of the game, so the next student can't resume or see it.
- **Online play:** room codes and device tokens now use the browser's cryptographic random generator.

October 2026 (update 11): a new name, Learn Around the Board (LAB)

- **Science Around the Board is now Learn Around the Board (LAB),** to reflect what it has become: a review game for any subject, not only the sciences. The rules, your question files and your results are unchanged.
- **New addresses:** the game is at https://hghezzi.github.io/Learn-Around-the-Board/, the guides are `LAB_Instructor_Guide.pdf` and `LAB_Student_Guide.pdf`, results files are named `lab_results_….csv`, and the question-writer skill is `lab-question-writer` (plugin: `/plugin marketplace add hghezzi/Learn-Around-the-Board`, then `/plugin install lab-question-writer@lab`).
- **Old files keep working:** encrypted `.lock` files made before the rename still open, and a results sheet set up with the old collector script keeps receiving results, so you don't need to change it. For a new sheet, use the new collector.

October 2026 (update 10): play against the bot, and Enter no longer skips the explanation

- **Solo players can play against the bot** (Easy, Medium or Hard) instead of chasing a goal. Its turns show its question, answer and explanation; **Skip ahead** speeds them up. Only the student's answers count. See *Playing against the bot*.
- **Fixed: pressing Enter skipped the explanation.** Pressing Enter to submit a typed answer (numbers and short answers) also pressed the next button, so the explanation vanished at once, and a right answer could even buy the tile without the player choosing. Enter now only submits, and the result's buttons ignore key presses for a moment. Keyboard users press Tab to reach **Continue**.
- **Results file:** against the bot, a second `GAME_RESULT` row with a new `bot` column (other games are unchanged).

October 2026 (update 9): license

- **The code is now under the PolyForm Noncommercial License 1.0.0**, a license written for software. The guides, question files and images stay under CC BY-NC-SA 4.0. Nothing changes for classes: teaching, learning and research use stays free, including for institutions that host their own copy. See *License and reuse*.
- **Your question files are yours**, whether you write them by hand or with the skill.

October 2026 (update 8): solo play

These change how solo games play and score. Games with 2–4 players are unchanged.

- **Your own tiles pay in solo.** Landing on a tile you own asks one of its questions; a right answer collects its rent from the bank, a wrong one costs $20. Until now, a solo player who bought well got fewer and fewer questions, and cash could only go down.
- **A solo goal:** reach a net-worth goal before time runs out ($5,000 in 60 minutes; see *Playing solo*). The board shows progress towards it, and the end screen shows the goal, the share of correct answers, the milestones and a personal best kept on that computer.
- **No chaos tokens in solo.**
- **Results file:** new `OWN_TILE_Q` and `SOLO_GOAL` rows and a `SOLO_RENT` transaction code (the columns are unchanged).

October 2026 (update 7): question-writer skill and file checker

- **A safer, more faithful question-writer skill.** It now follows your materials closely (notation, examples, conventions), raises errors it finds in your notes, checks every

number and code snippet, and keeps to ground rules on student data, copyright, live exams, hidden instructions in uploaded files and sensitive topics. See *Generating questions with Claude*.

- **New file checks:** answer cues for each game in a file and for the milestone exams; cells that a spreadsheet would turn into a formula or strip of an apostrophe; select-all questions that always have the same number of correct options; short answers the typo rule could confuse (absorption/adsorption); and a Google Form or Sheet link given instead of the collector's link (now a red item).
- **"Results are sent to"** is the new last line of the check: it says where results go and whether names are required.
- **Short answers: Roman numerals count like numbers.** A typo is never forgiven in a Roman numeral, so "Type II error" is no longer accepted for "Type I error" (nor Photosystem I for II).
- **Numeric answers accept a unit:** "12 kg", "0.5 mL" or "25 °C" counts as the number.
- **Required names now hold back Download too.** With `ask_names` on (the default when results are sent or emailed), **Send**, **Email** and **Download** stay disabled until every player's names or IDs are typed.
- **Updated counts table** for 45-, 60- and 90-minute sessions (see *How many questions?*).
- **Demo file:** nine cells were retyped so they survive a round trip through Excel or Google Sheets. No question or answer changed.
- **End game and Exit session ask first.** Both now open a yes/no confirmation (**Yes, end the game** / **No, keep playing**, and **Yes, leave** / **No, stay in the game**), so a stray click can't end or throw away a game.
- **The question-writer skill is now a Claude Code plugin:** `/plugin marketplace add hghezzi/Learn-Around-the-Board`, then `/plugin install lab-question-writer@lab`. The zip for claude.ai is updated too.
- **A privacy warning before you share materials with Claude:** remove student data and anything confidential first, and check your Claude account's privacy settings (see *Generating questions with Claude*).
- **The question-writer skill asks about confidence sliders:** whether you want them at all, and which skills they cover (it drafts 4–5 from your learning objectives for you to keep, edit or drop).

October 2026 (update 6): online play (beta)

- **Players on their own devices.** On the setup screen, choose *On their own devices (beta)*: the host computer opens a room with a code and a link, players join from any

computer, tablet or phone, answer their surveys at the same time and play their turns on their own device. Refreshes reconnect by themselves, and a host refresh resumes the same room. Playing on one computer is unchanged and stays the default. See Online play (beta).
- The start page has **Join an online game**, and the privacy notice explains the connection service online play uses.

October 2026 (update 5)

- **Starting cash depends on the number of players:** $2,500 solo, $2,000 each with 2 players, $1,500 with 3 and $1,250 with 4. Fewer players get more turns and more tiles each, so they need more cash; I chose the amounts with a simulation of 3,000 games per setting so that 2, 3 and 4 players feel a similar squeeze.
- **Gentler fees:** landing on a rival's milestone costs $250 again (it was $625), and a core tile's rent now grows with the owner's collection: $50 for each of the 4 core tiles they hold, up to $200.
- **"Player" everywhere in the game:** the game now says *Red Player*, *Solo Player*, "you pay…" and so on, instead of "team". A group of students can still share one player. The `TEAM_INFO` code in the results file is unchanged.

October 2026 (update 4): gameplay changes. These change how the game plays and scores.

- **Unseen questions first, missed ones later.** The game now asks questions nobody has seen before repeating any, and a question answered wrongly comes back about six turns later (spaced retrieval). Exams, chaos challenges and the Rescue Quiz follow the same rule.
- **A tighter economy.** Teams start with $1,500 (was $2,500) and every rent is 2.5 times higher: property rent is 50% of the price, core rent is $300 and a rival's milestone fee is $625. Debt, liquidation and the Rescue Quiz now happen in real games.
- **Net worth counts only money spent.** A tile counts at what its owner paid for it, and each upgrade counts once, as it was charged (it used to count once per tile, so a $160 upgrade added $480). Every sale during liquidation returns exactly half of what was paid.
- **Chaos challenges:** a complete set (all three tiles, upgraded or not) can no longer be challenged; using a token ends the turn, and the token is spent whether or not the steal succeeds; the Challenge button stays off until the team can pay the price.
- **Fairer short answers.** One typo is now forgiven only in answers of 8 or more characters, never in the first letter or a number, so `alkene` is no longer accepted for

`alkane`, nor `methanol` for `ethanol`.

- **Quick rules.** A new game opens with a short **How to play** pop-up; teams can reopen it from the board, and the start page has a **Quick rules** link.
- **"Players" instead of "teams" on the setup screen:** choose Solo, 2, 3 or 4 players (each player can still be a small team).
- **CSV:** survey rows' `selectedIndex` is now the option number in your file (1–4, like `correctIndex`) instead of the shuffled position on screen, and `TRANSACTION` rows have a `timestamp`.
- **New demo images.** The demo's five figures were redrawn from made-up data so they can be shared freely; the older hosted images are no longer served. If your own question file used those hosted file names, upload the images with **Optional: upload images** or share them with an image-folder link.
- **Automatic deploys:** every change merged into the project goes live after all checks pass.

October 2026 (update 3)

- **Snappier turns.** The pawn moves a little faster and its dialog opens sooner (about half a second saved per roll, about 20 seconds over a 45-minute game). The rules are unchanged.
- **Stronger `.lock` files.** The encryptor now uses modern browser encryption (PBKDF2 and AES-256-GCM) and asks for a password of at least 8 characters. Your existing `.lock` files still open; re-encrypt them to benefit.
- **Everyone gets fixes promptly.** An open game checks for a new version regularly. On an empty start page it reloads by itself; during a session it shows a **Reload** notice and never interrupts the game.
- **Results collector, version 2.** The Google Sheet script now rejects oversized or malformed submissions. If you set it up before, paste in the new script and create a new version of the deployment (the URL stays the same).
- **Privacy:** changing your mind about analytics switches it off at once and deletes its cookies.
- **The file checker now looks for answers that give themselves away.** When the correct option is usually the longest one, students can win without knowing anything. The checker reports how often "always pick the longest option" (or the shortest, or the one that repeats the question's words) would be right, compared with chance, and lists questions whose correct answer is much longer than the others. It also flags "all of the above", absolute words such as "always" or "never" that appear only in wrong options,

survey questions that repeat a board question, and spreadsheet errors such as `#NAME?`. See *Checking your file*.
- **Revised demo and statistics questions.** In the 16S demo, picking the longest option used to be right 69% of the time; it is now 21% (chance is 25%). Several questions were corrected (for example, QIIME 2 commands and flags), two questions whose options a spreadsheet had turned into `#NAME?` were restored, a few questions now use the numeric, ordering, select-all and short-answer formats, and there are six new wildcards.
- **The question-writer skill** now checks every batch for these cues before building the file.

October 2026 (documentation)

- **A Student Guide**, on the web and as a printable PDF, explains the rules and how to hand in results in a few minutes. Share it with your class.
- **This guide** now starts with a Quick start and adds the scoring numbers at a glance, how many questions to write, spreadsheet pitfalls, ideas for analysing and grading results, and examples for other subjects. It replaces the older separate documentation pages.

October 2026 (update 2)

- **Results can go straight to you.** Students type their names or student IDs on the end screen, then click **Send results to instructor** (an optional Google Sheet you set up once), **Email results to instructor**, or **Download results**. See Collecting results.
- **Shareable game links.** Paste a link to your question file, for example a published Google Sheet, and give students one link: the questions load by themselves. See Sharing the game with students.
- **Autosave.** An accidental refresh now offers **Resume your game?** instead of losing the session.
- **Install as an app and play offline** after the first visit.
- **Password dialog** for encrypted `.lock` files, with a clear message when the password is wrong.
- **Wildcard tiles** (formerly "Chance"), a **lap bonus** for passing START and a new board colour. The rules are unchanged.
- **Demo fixes:** every team now gets its survey questions, whatever order you choose the topic and the number of teams in (teams 3 and 4 used to get none); the demo now includes questions with images; image notices list only images that truly can't be found.

- **The in-game "Export CSV" button is gone.** It saved only the game log, so students could hand in an incomplete file. The end screen exports everything, now as `lab_results_<topic>_<module>_<date>.csv` with a `TEAM_INFO` row per team.
- **"Exit session" asks for confirmation.**
- **The question-writer skill** now offers every question format (defaulting to all of them) and sets up how results reach you.

October 2026 (update)

- **Subject-neutral wording.** Everything the game says itself now fits any field: Chance tiles instead of "Lab Mishaps", **Buy** instead of "Publish", **Upgrades** instead of "Lab manager", a **Rescue Quiz** instead of an "Emergency Grant", and so on. Your own themes, questions and chance-card text are shown exactly as you write them, so a science course can still have lab-themed events.
- If your file has no `mishap` rows, the built-in chance cards are now general study-life events with learning tips.

October 2026

- **Works in dark mode.** Questions used to be invisible when a browser was set to dark mode. The game now follows each device’s light or dark setting, and both are readable.
- **A new look:** a redesigned board with coloured theme bands, animated dice, team symbols (●▲■◆) so teams are not told apart by colour alone, and clearer answer feedback.
- **New ways to win:** last team standing, or the highest net worth when time runs out. An optional session timer is set on the setup screen.
- **Bankruptcy has consequences:** each team gets one Rescue Quiz. Fail it, or go bankrupt again, and the team is eliminated.
- **New question formats:** select-all-that-apply, numeric (with tolerance), ordering and short answer, alongside multiple choice and true/false.
- **Answer options are shuffled** every time a question appears, and wrong answers now show the correct one.
- **File checker:** when a question file is loaded, the game lists any problems before you start.
- **Images are optional** and missing ones are simply hidden.
- **Chaos challenges and rescue quizzes use your own questions** (previously they used built-in bioinformatics questions).
- **Chance cards (`mishap` rows) pay or cost the amount written on the card**, for example *(-$100)*.

- **Richer CSV export:** one row per answered question, with the response, the correct answer and the final standings.
- **Question-writer for Claude:** a skill that interviews you and writes a complete, validated question file.
- **A second example game:** Intro Statistics, to show the engine outside biology.
- **Privacy:** anonymous usage analytics load only if a visitor opts in.

The March 2026 edition of this guide is archived as a PDF.